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(54) **SEMICONDUCTOR DEVICE**
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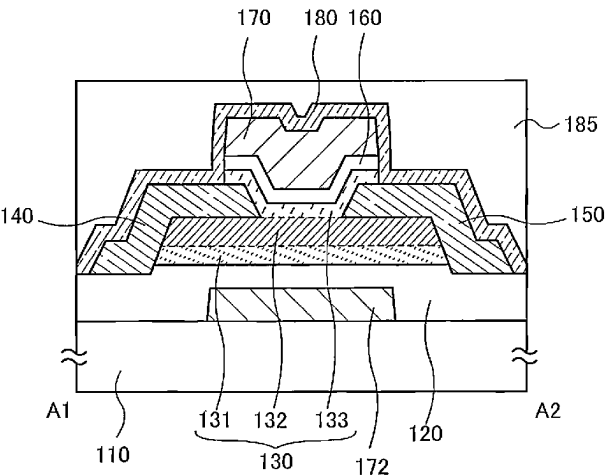
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(57) **ABSTRACT**
A semiconductor device having a structure which can prevent a decrease in electrical characteristics due to miniaturization is provided. The semiconductor device includes, over an insulating surface, a stack in which a first oxide semiconductor layer and a second oxide semiconductor layer are sequentially formed, and a third oxide semiconductor layer covering part of a surface of the stack. The third oxide semiconductor layer includes a first layer in contact with the stack and a second layer over the first layer. The first layer includes a microcrystalline layer, and the second layer includes a crystalline layer in which c-axes are aligned in a direction perpendicular to a surface of the first layer.

7 Claims, 15 Drawing Sheets



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continuation of application No. 17/564,518, filed on Dec. 29, 2021, now Pat. No. 11,646,380, which is a continuation of application No. 16/923,160, filed on Jul. 8, 2020, now Pat. No. 11,217,704, which is a continuation of application No. 16/558,601, filed on Sep. 3, 2019, now Pat. No. 10,720,532, which is a continuation of application No. 16/126,348, filed on Sep. 10, 2018, now Pat. No. 10,411,136, which is a continuation of application No. 15/828,759, filed on Dec. 1, 2017, now Pat. No. 10,128,384, which is a continuation of application No. 15/246,927, filed on Aug. 25, 2016, now Pat. No. 9,837,552, which is a continuation of application No. 15/049,554, filed on Feb. 22, 2016, now Pat. No. 9,431,547, which is a continuation of application No. 14/276,294, filed on May 13, 2014, now Pat. No. 9,281,408.

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FIG. 1A

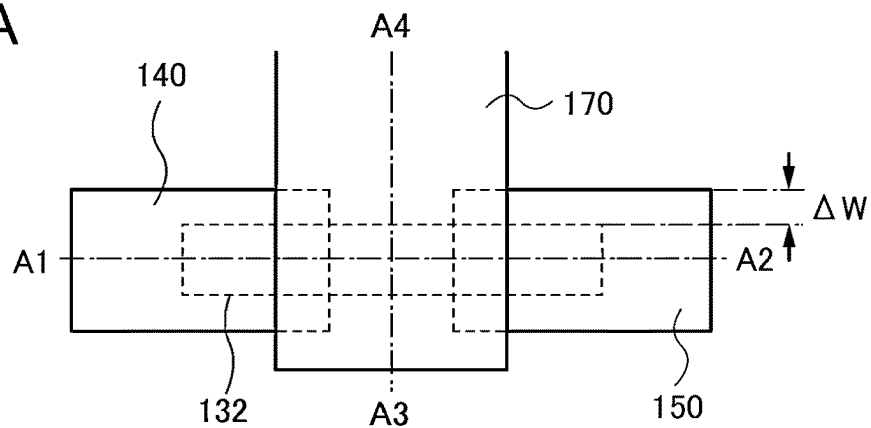


FIG. 1B

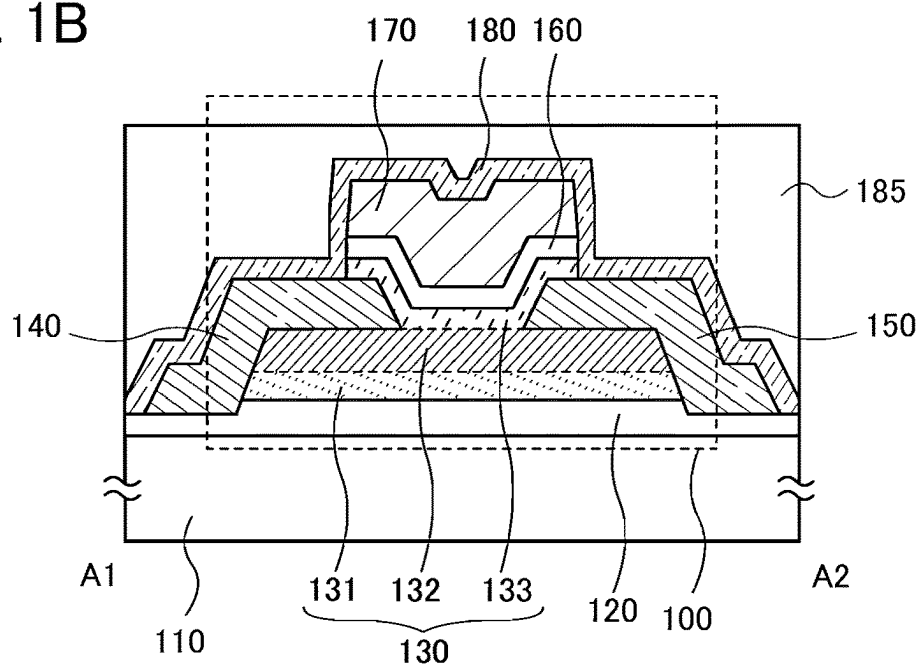
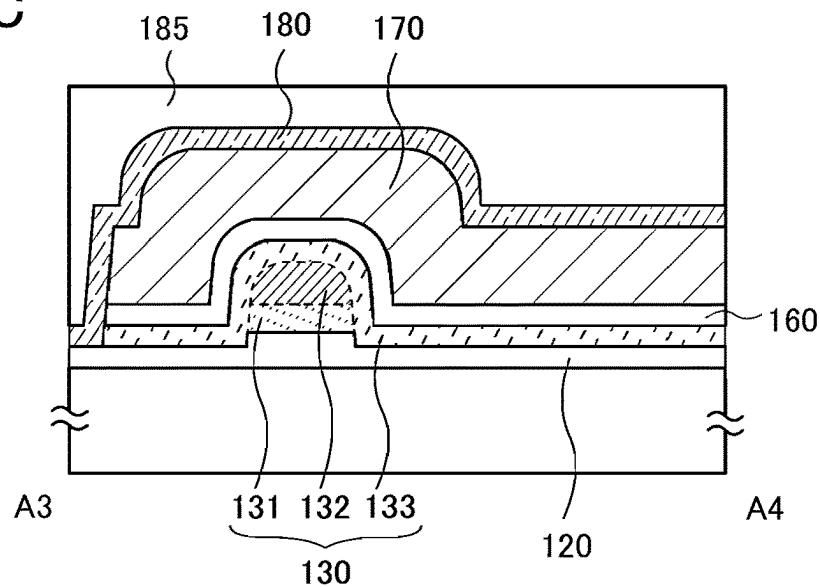


FIG. 1C



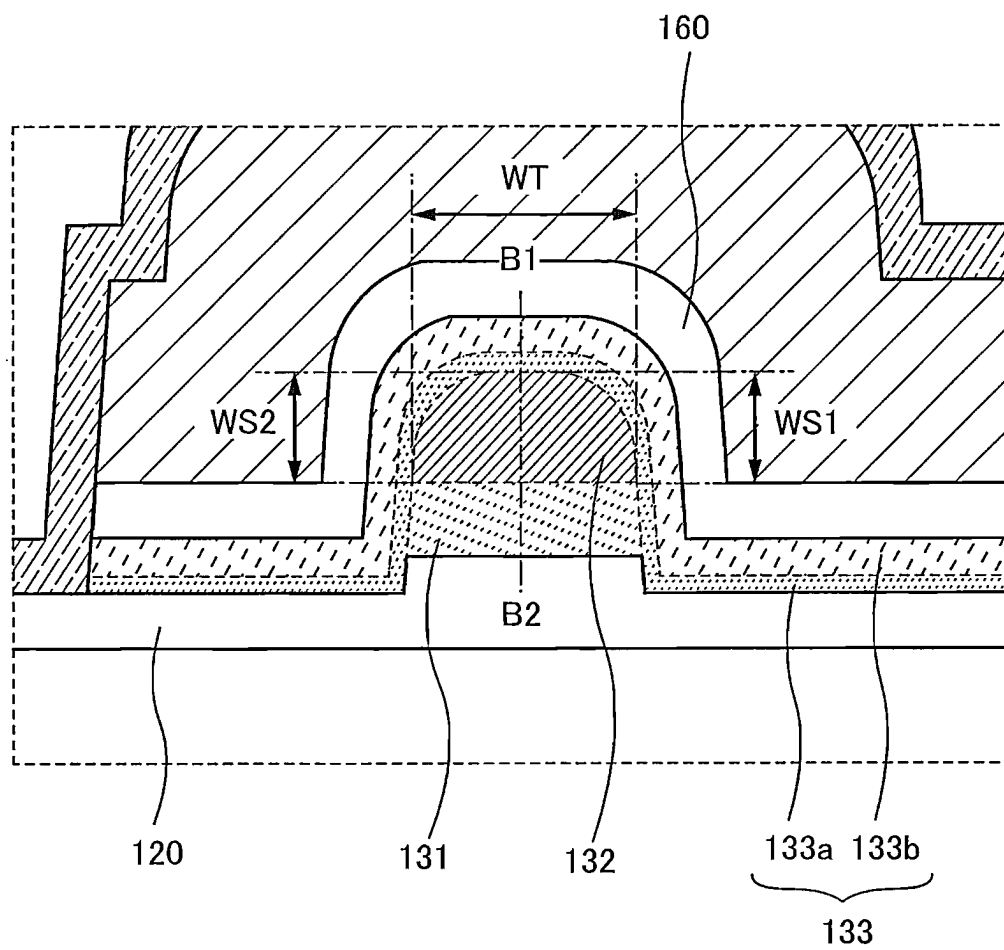


FIG. 3

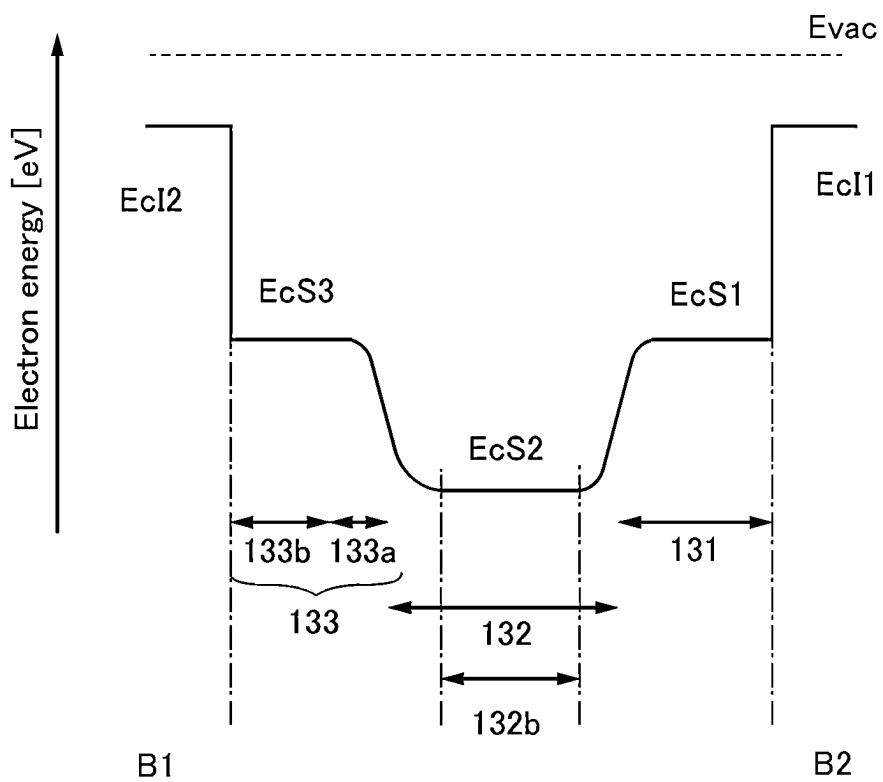


FIG. 4A

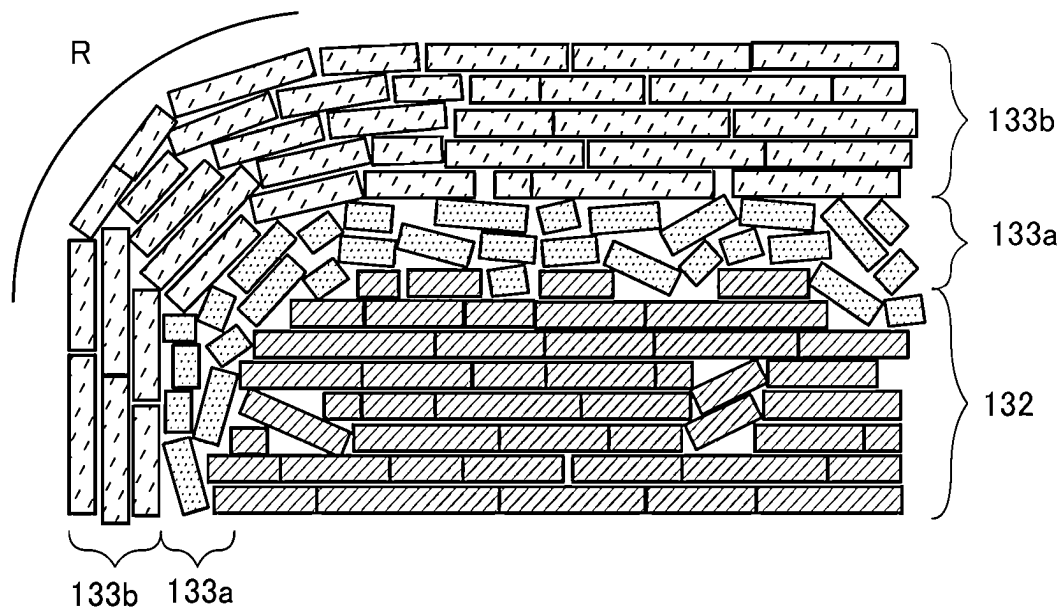


FIG. 4B

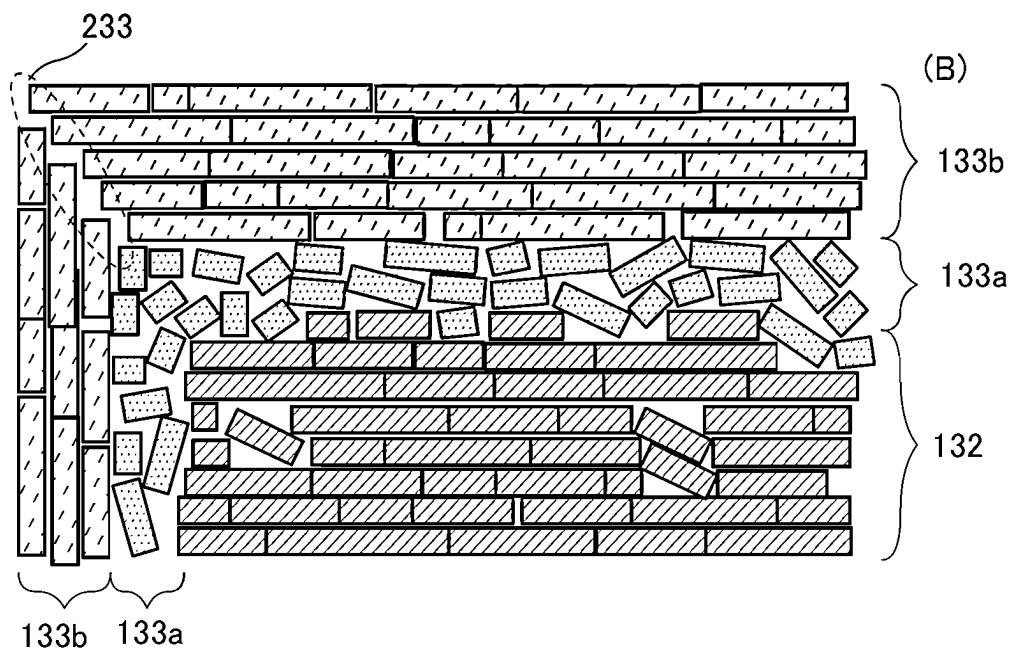


FIG. 5

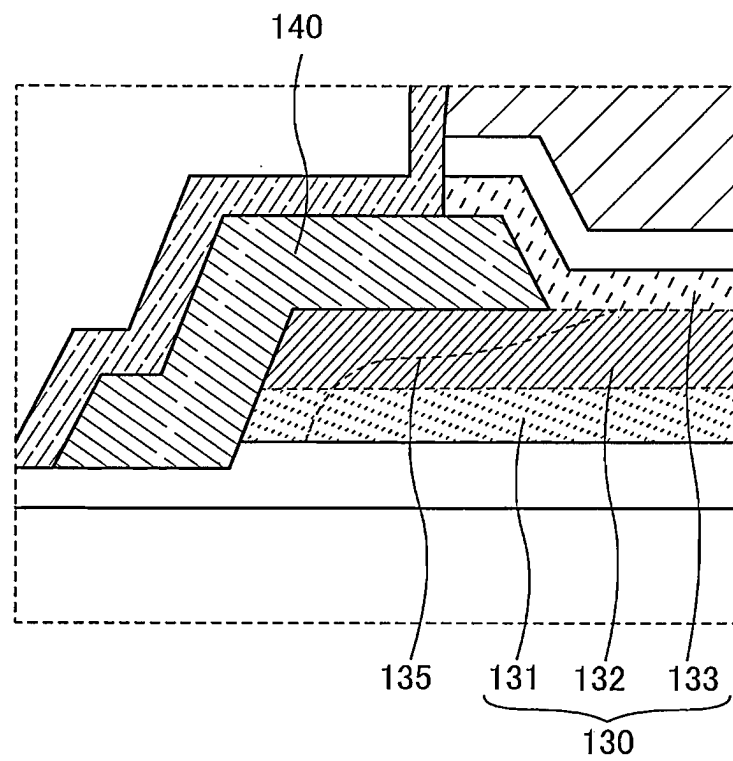


FIG. 6

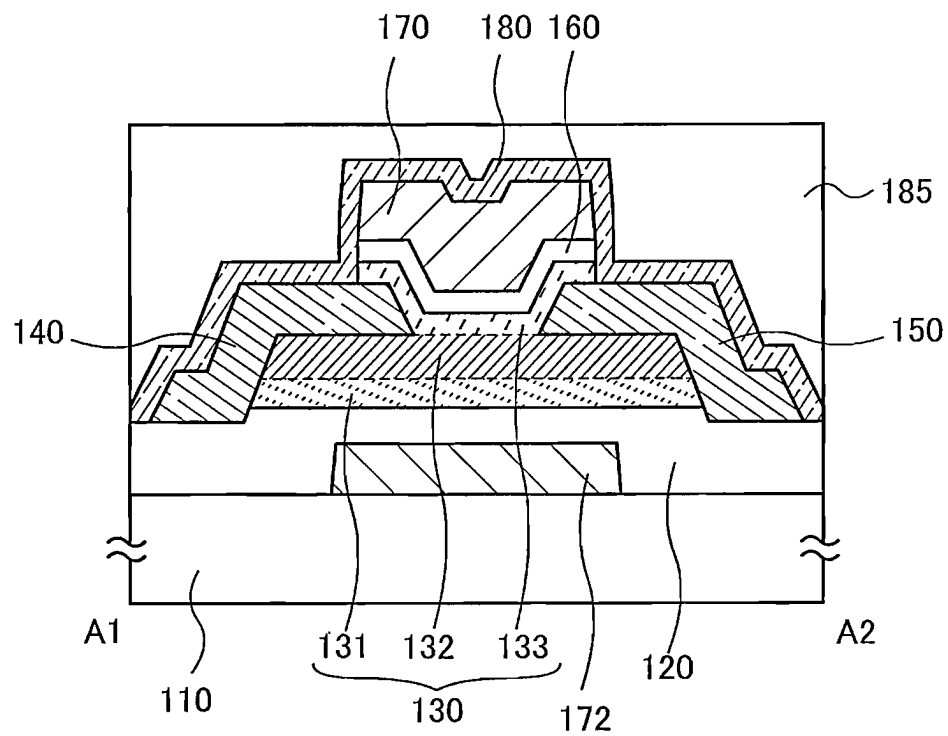


FIG. 7A

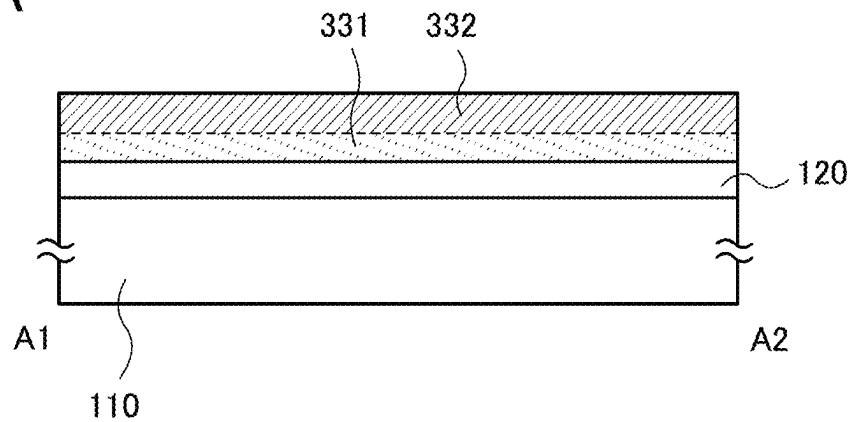


FIG. 7B

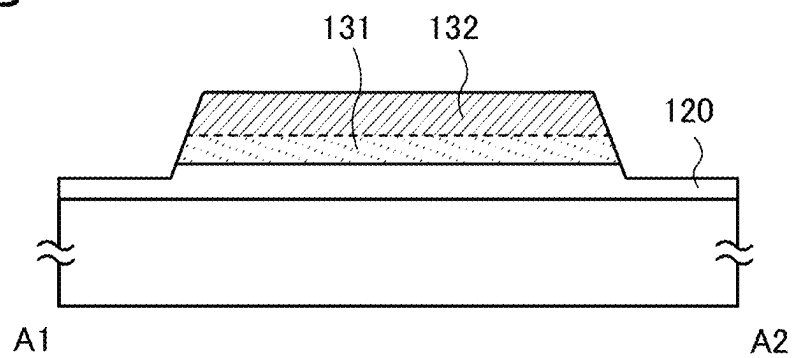


FIG. 7C

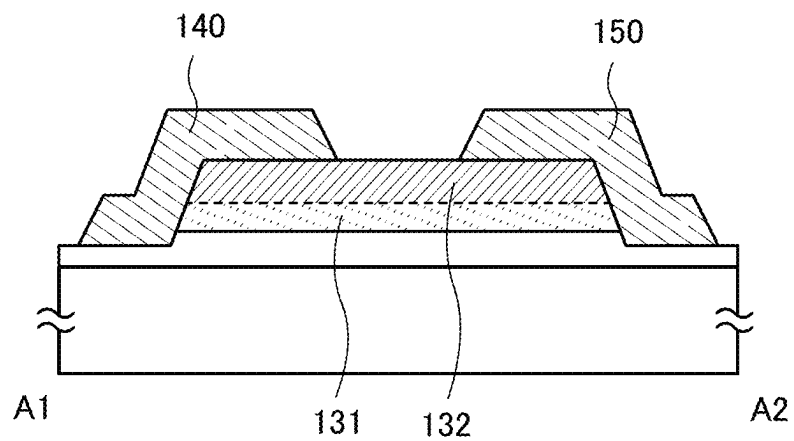


FIG. 8A

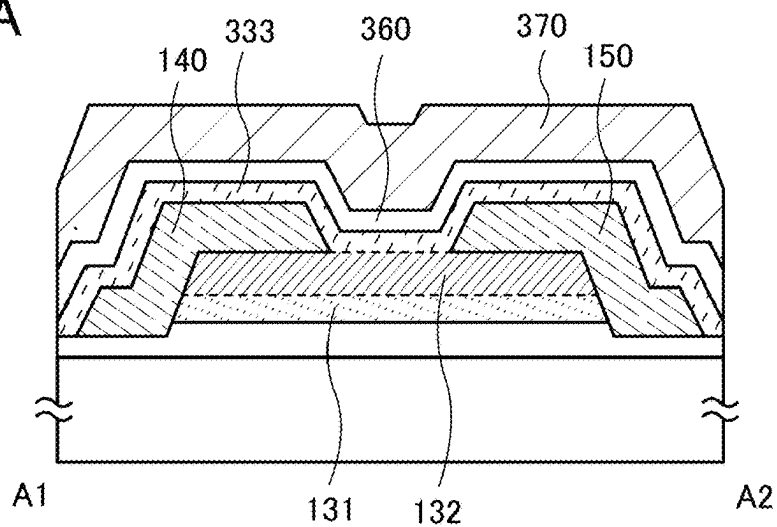


FIG. 8B

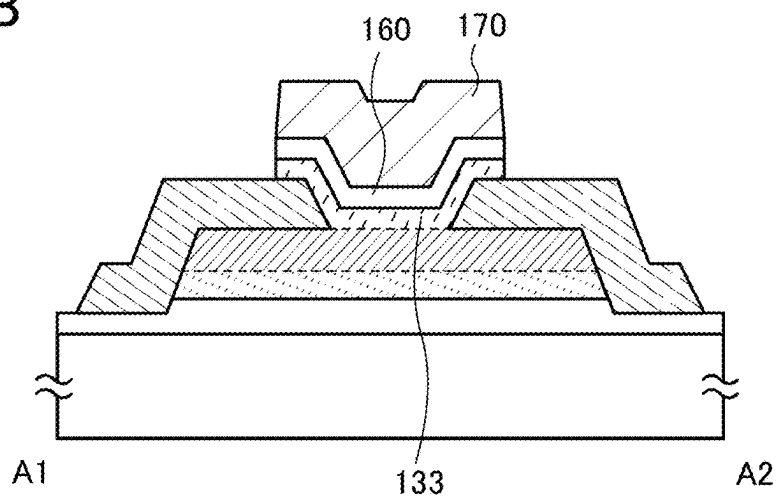


FIG. 8C

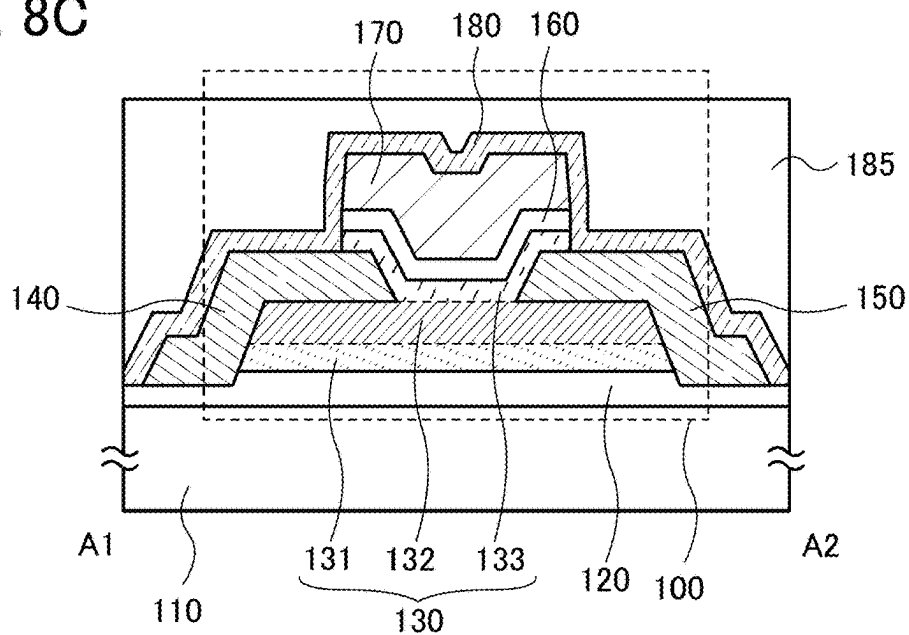


FIG. 9A

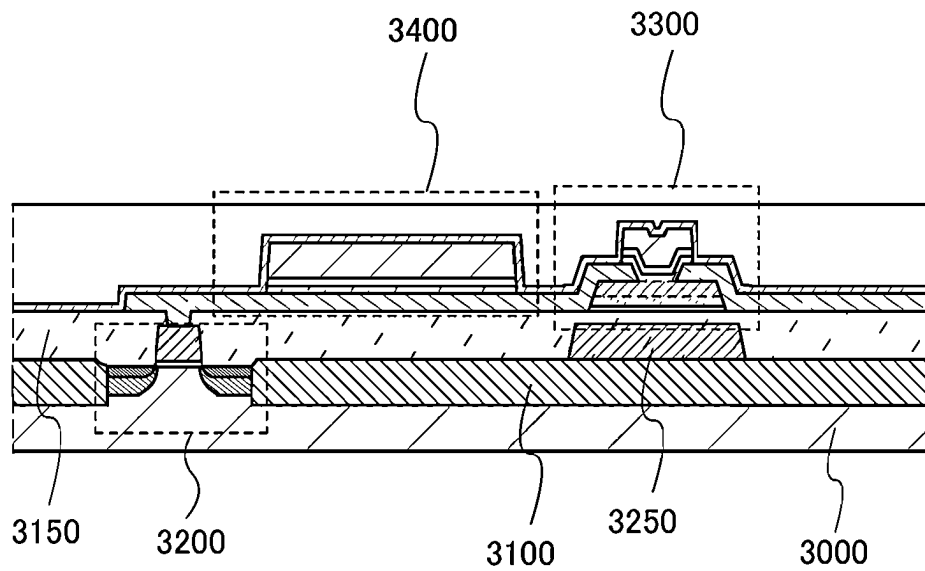


FIG. 9B

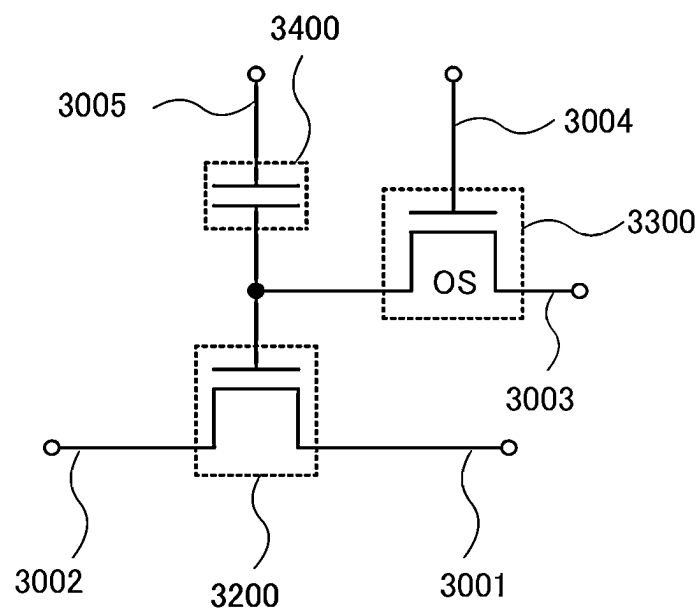


FIG. 10

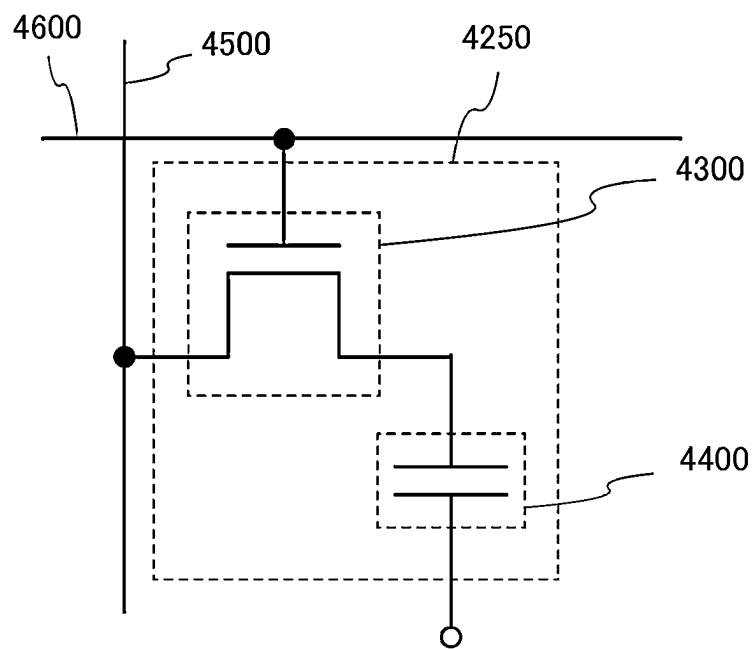


FIG. 11A

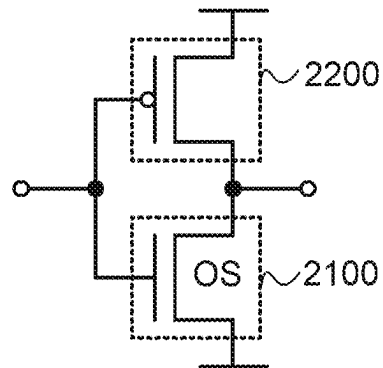


FIG. 11B

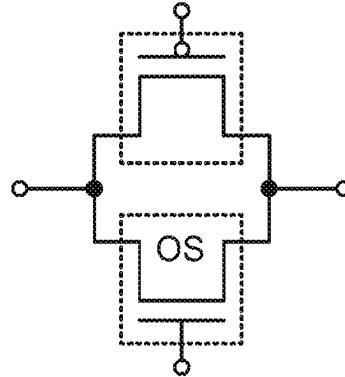


FIG. 11C

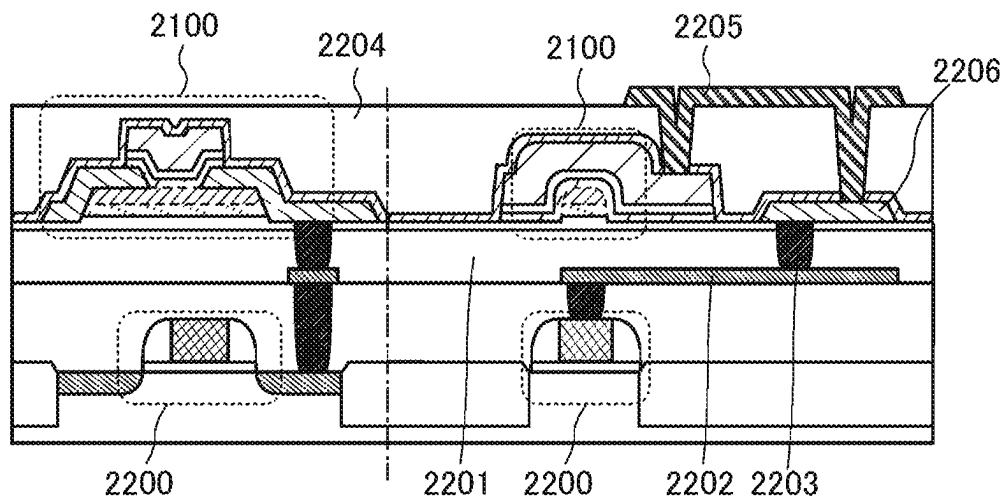


FIG. 11D

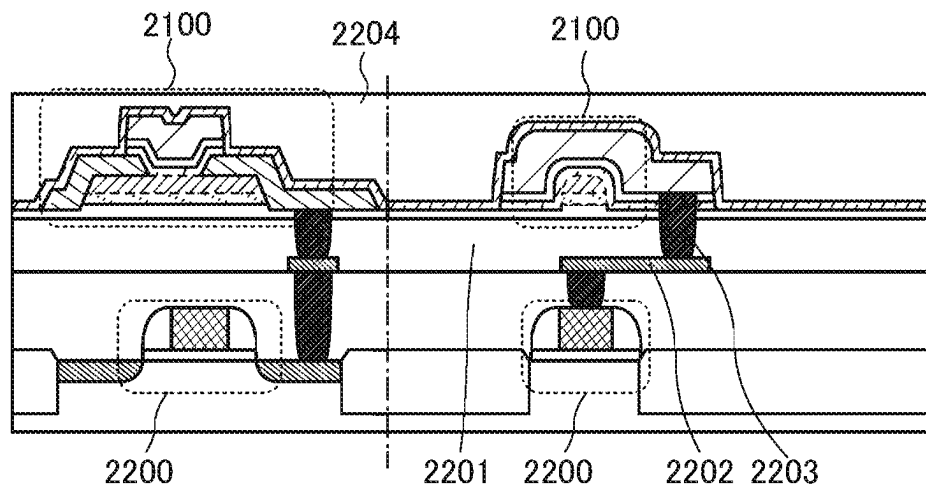


FIG. 12

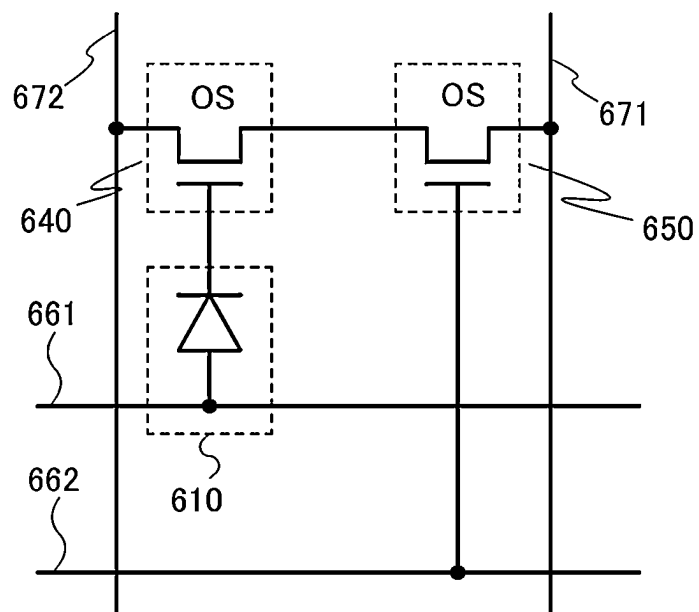


FIG. 13A

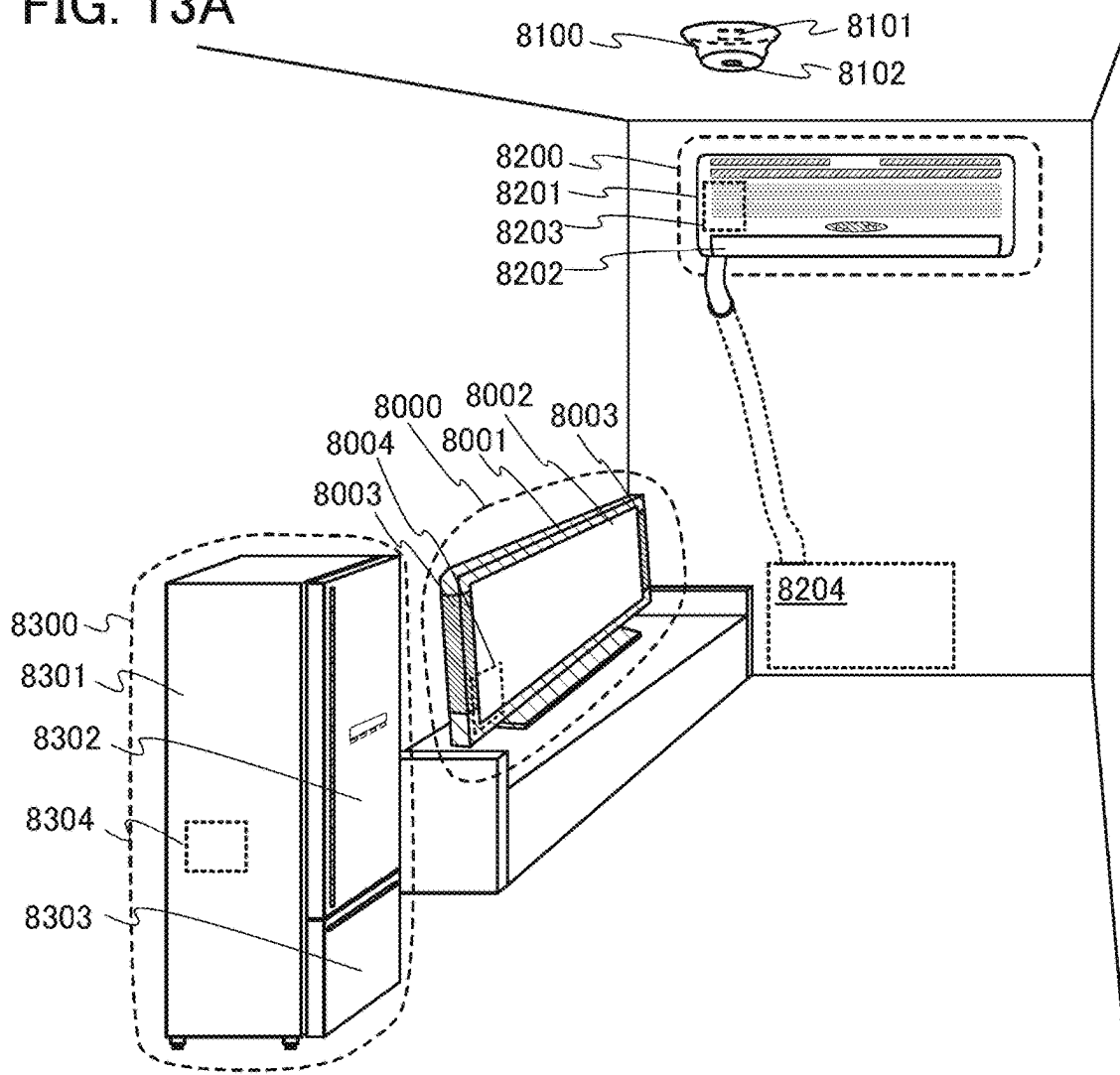


FIG. 13B

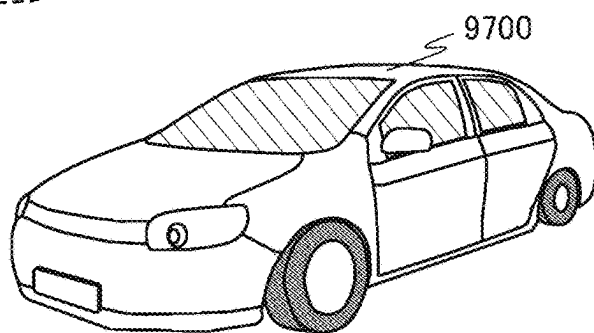


FIG. 13C

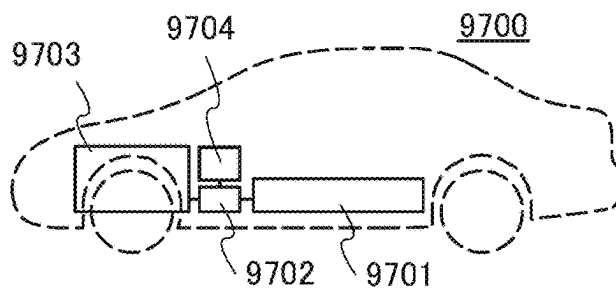


FIG. 14

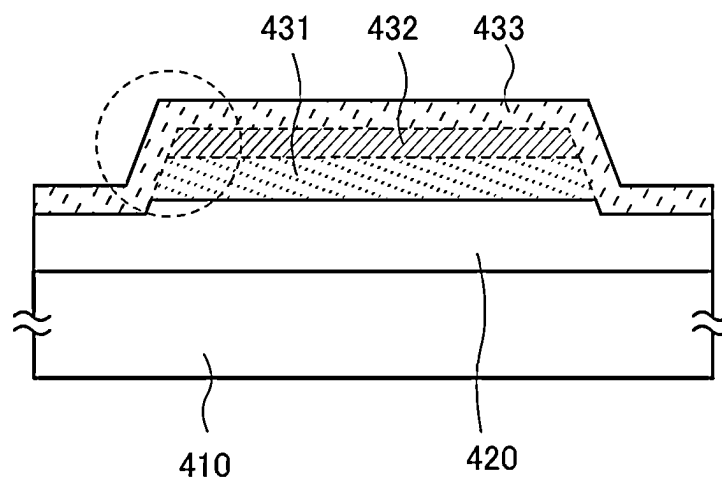


FIG. 15A

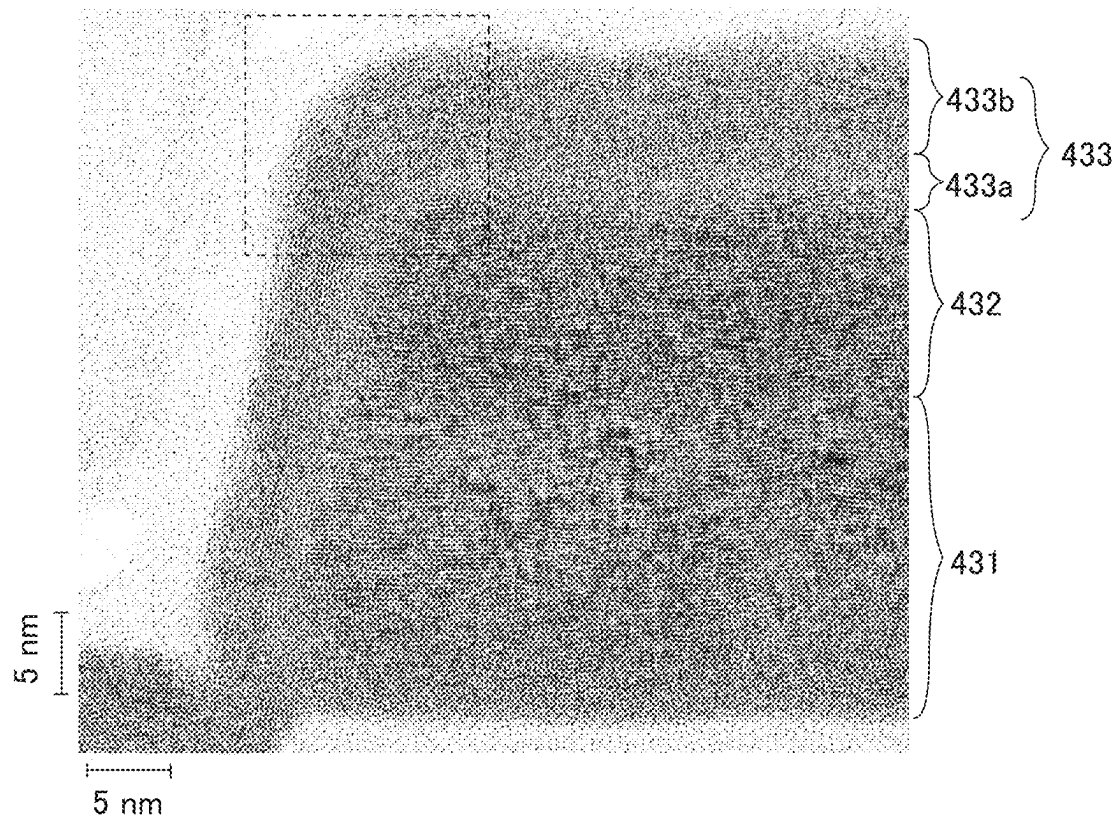
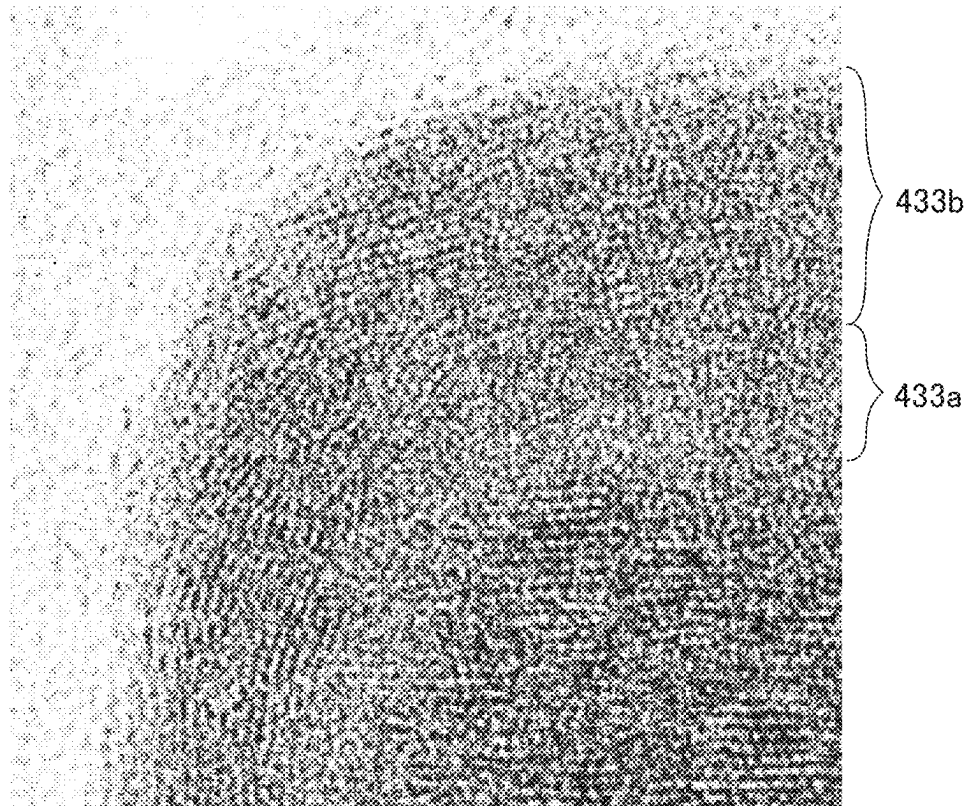


FIG. 15B



1

SEMICONDUCTOR DEVICE

BACKGROUND OF THE INVENTION

1. Field of the Invention

One embodiment of the present invention relates to a semiconductor device including an oxide semiconductor.

Note that one embodiment of the present invention is not limited to the above technical field. The technical field of one embodiment of the invention disclosed in this specification and the like relates to an object, a method, or a manufacturing method. In addition, one embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter. Specifically, examples of the technical field of one embodiment of the present invention disclosed in this specification include a semiconductor device, a display device, a liquid crystal display device, a light-emitting device, a lighting device, a power storage device, a storage device, a method for driving any of them, and a method for manufacturing any of them.

In this specification and the like, a semiconductor device generally means a device that can function by utilizing semiconductor characteristics. A transistor and a semiconductor circuit are embodiments of semiconductor devices. In some cases, a storage device, a display device, or an electronic device includes a semiconductor device.

2. Description of the Related Art

Attention has been focused on a technique for forming a transistor using a semiconductor thin film formed over a substrate having an insulating surface (also referred to as a thin film transistor (TFT)). The transistor is used in a wide range of electronic devices such as an integrated circuit (IC) and an image display device (display device). A silicon-based semiconductor material is widely known as a material for a semiconductor thin film applicable to a transistor. As another example, an oxide semiconductor has been attracting attention.

For example, a transistor whose active layer includes an amorphous oxide semiconductor containing indium (In), gallium (Ga), and zinc (Zn) is disclosed in Patent Document 1.

REFERENCE

Patent Document

[Patent Document 1] Japanese Published Patent Application No. 2006-165528

SUMMARY OF THE INVENTION

High integration of an integrated circuit requires miniaturization of a transistor. However, it is known that miniaturization of a transistor causes deterioration of or variations in the electrical characteristics of the transistor. This means that miniaturization of a transistor is likely to decrease in the yield of an integrated circuit.

Thus, one object of one embodiment of the present invention is to provide a semiconductor device in which deterioration of electrical characteristics which becomes more noticeable as the transistor is miniaturized can be suppressed. Another object is to provide a semiconductor device having a structure with which a decrease in a yield due to miniaturization can be suppressed. Another object is

2

to provide a semiconductor device having a high degree of integration. Another object is to provide a semiconductor device in which deterioration of on-state current characteristics is reduced. Another object is to provide a semiconductor device with low power consumption. Another object is to provide a semiconductor device with high reliability. Another object is to provide a semiconductor device which can retain data even when power supply is stopped. Another object is to provide a novel semiconductor device.

Note that the descriptions of these objects do not disturb the existence of other objects. Note that in one embodiment of the present invention, there is no need to achieve all the objects. Other objects are apparent from and can be derived from the description of the specification, the drawings, the claims, and the like.

One embodiment of the present invention relates to a semiconductor device having a stack including oxide semiconductor layers.

One embodiment of the present invention is a semiconductor device including, over an insulating surface, a stack in which a first oxide semiconductor layer and a second oxide semiconductor layer are sequentially formed, and a third oxide semiconductor layer. The third oxide semiconductor layer covers part of a first side surface, part of a top surface, and part of a second side surface opposite to the first side surface of the stack. The third oxide semiconductor layer includes a first layer in contact with the stack, and a second layer over the first layer. The first layer includes a microcrystalline layer, and the second layer includes a crystalline layer in which c-axes are aligned in a direction perpendicular to a surface of the first layer.

Another embodiment of the present invention is a semiconductor device including, over an insulating surface, a stack in which a first oxide semiconductor layer and a second oxide semiconductor layer are sequentially formed; a source electrode layer and a drain electrode layer each partly in contact with the stack; a third oxide semiconductor layer partly in contact with each of the insulating surface, the stack, the source electrode layer, and the drain electrode layer; a gate insulating film over the third oxide semiconductor layer; a gate electrode layer over the gate insulating film; and an insulating layer over the source electrode layer, the drain electrode layer, and the gate electrode layer. The third oxide semiconductor layer includes a first layer in contact with the stack, and a second layer over the first layer. The first layer includes a microcrystalline layer, and the second layer includes a crystalline layer in which c-axes are aligned in a direction perpendicular to a surface of the first layer.

Note that in this specification and the like, ordinal numbers such as “first” and “second” are used in order to avoid confusion among components and do not limit the components numerically.

The first oxide semiconductor layer preferably includes a crystalline layer in which c-axes are aligned in a direction perpendicular to the insulating surface. The second oxide semiconductor layer preferably includes a crystalline layer in which c-axes are aligned in a direction perpendicular to a top surface of the first oxide semiconductor layer.

Further, a surface of the second oxide semiconductor layer is preferably curved in a region where the stack is in contact with the third oxide semiconductor layer.

Further, a conduction band minimum of the first oxide semiconductor layer and a conduction band minimum of the third oxide semiconductor layer are preferably closer to a

vacuum level than a conduction band minimum of the second oxide semiconductor layer by 0.05 eV or more and 2 eV or less.

It is preferable that the first to third oxide semiconductor layers each include an In-M-Zn oxide (M is Al, Ti, Ga, Y, Zr, La, Ce, Nd, or Hf), and that an atomic ratio of M with respect to In in each of the first and third oxide semiconductor layers be higher than an atomic ratio of M with respect to In in the second oxide semiconductor layer.

According to one embodiment of the present invention, any of the following effects can be achieved: to provide a semiconductor device in which deterioration of electrical characteristics which becomes more noticeable as the semiconductor device is miniaturized can be suppressed, to provide a semiconductor device that can be miniaturized in a simple process, to provide a semiconductor device having a structure with which a decrease in a yield due to miniaturization can be suppressed, to provide a semiconductor device having a high degree of integration, to provide a semiconductor device in which deterioration of on-state current characteristics is reduced, to provide a semiconductor device with low power consumption, to provide a semiconductor device with high reliability, to provide a semiconductor device which can retain data even when power supply is stopped, and to provide a novel semiconductor device.

Note that the descriptions of these effects do not disturb the existence of other effects. In one embodiment of the present invention, there is no need to obtain all the effects. Other effects are apparent from and can be derived from the description of the specification, the drawings, the claims, and the like.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A to 1C are a top view and cross-sectional views of a transistor.

FIG. 2 is a cross-sectional view of a transistor.

FIG. 3 illustrates a band structure of oxide semiconductor layers.

FIGS. 4A and 4B each illustrate a crystal structure of part of a stack including oxide semiconductor layers.

FIG. 5 is an enlarged cross-sectional view of a transistor.

FIG. 6 is a cross-sectional view of a transistor.

FIGS. 7A to 7C illustrate a method for manufacturing a transistor.

FIGS. 8A to 8C illustrate a method for manufacturing a transistor.

FIGS. 9A and 9B are a cross-sectional view and a circuit diagram of a semiconductor device.

FIG. 10 is a circuit diagram of a semiconductor device.

FIGS. 11A and 11B are each a circuit diagram of a semiconductor device and FIGS. 11C and 11D are each a cross-sectional view of a semiconductor device.

FIG. 12 is a circuit diagram of a semiconductor device.

FIGS. 13A to 13C illustrate electronic devices in which semiconductor devices can be used.

FIG. 14 is a cross-sectional view of a sample for observing a stacked structure of oxide semiconductor layers.

FIGS. 15A and 15B are each a cross-sectional TEM photograph of oxide semiconductor layers.

DETAILED DESCRIPTION OF THE INVENTION

Embodiments are described in detail with reference to the drawings. Note that the present invention is not limited to

the following description and it is readily appreciated by those skilled in the art that modes and details can be modified in various ways without departing from the spirit and the scope of the present invention. Therefore, the present invention should not be limited to the descriptions of the embodiments below. Note that in structures of the present invention described below, the same portions or portions having similar functions are denoted by the same reference numerals in different drawings, and description thereof is omitted in some cases.

Note that in this specification and the like, when it is explicitly described that X and Y are connected, the case where X and Y are electrically connected, the case where X and Y are functionally connected, and the case where X and Y are directly connected are included therein. Here, each of X and Y denotes an object (e.g., a device, an element, a circuit, a wiring, an electrode, a terminal, a conductive film, a layer, or the like). Accordingly, a connection relation other than connection relations shown in the drawings and texts is also included, without being limited to a predetermined connection relation, for example, a connection relation shown in the drawings and texts.

In the case where X and Y are electrically connected, one or more elements (e.g., a switch, a transistor, a capacitor, an inductor, a resistor, a diode, a display element, a light-emitting element, and a load) that enable an electrical connection between X and Y can be connected between X and Y, for example. Note that the switch is controlled to be turned on or off. That is, the switch has a function of determining whether current flows or not by being turned on or off (becoming an on state and an off state). Alternatively, the switch has a function of selecting and changing a current path.

In the case where X and Y are functionally connected, one or more circuits (e.g., a logic circuit such as an inverter, a NAND circuit, or a NOR circuit; a signal converter circuit such as a DA converter circuit, an AD converter circuit, or a gamma correction circuit; a potential level converter circuit such as a power supply circuit (e.g., a step-up circuit or a step-down circuit) or a level shifter circuit for changing the potential level of a signal; a voltage source; a current source; a switching circuit; an amplifier circuit such as a circuit that can increase signal amplitude, the amount of current, or the like, an operational amplifier, a differential amplifier circuit, a source follower circuit, or a buffer circuit; a signal generation circuit; a storage circuit; and a control circuit) that enable a functional connection between X and Y can be connected between X and Y, for example. Note that for example, in the case where a signal output from X is transmitted to Y even when another circuit is interposed between X and Y, X and Y are functionally connected.

Note that when it is explicitly described that X and Y are connected, the case where X and Y are electrically connected (i.e., the case where X and Y are connected with another element or another circuit provided therebetween), the case where X and Y are functionally connected (i.e., the case where X and Y are functionally connected with another circuit provided therebetween), and the case where X and Y are directly connected (i.e., the case where X and Y are connected without another element or another circuit provided therebetween) are included therein. That is, when it is explicitly described that “X and Y are electrically connected”, the description is the same as the case where it is explicitly only described that “X and Y are connected”.

Even when independent components are electrically connected to each other in a circuit diagram, one component has functions of a plurality of components in some cases. For

5

example, when part of a wiring also functions as an electrode, one conductive film functions as the wiring and the electrode. Thus, an "electrical connection" in this specification includes in its category such a case where one conductive film has functions of a plurality of components.

Note that, for example, the case where a source (or a first terminal or the like) of a transistor is electrically connected to X through (or not through) Z1 and a drain (or a second terminal or the like) of the transistor is electrically connected to Y through (or not through) Z2, or the case where a source (or a first terminal or the like) of a transistor is directly connected to one part of Z1 and another part of Z1 is directly connected to X while a drain (or a second terminal or the like) of the transistor is directly connected to one part of Z2 and another part of Z2 is directly connected to Y, can be expressed by using any of the following expressions.

The expressions include, for example, "X, Y, a source (or a first terminal or the like) of a transistor, and a drain (or a second terminal or the like) of the transistor are electrically connected to each other, and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are electrically connected to each other in this order", "a source (or a first terminal or the like) of a transistor is electrically connected to X, a drain (or a second terminal or the like) of the transistor is electrically connected to Y, and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are electrically connected to each other in this order", and "X is electrically connected to Y through a source (or a first terminal or the like) and a drain (or a second terminal or the like) of a transistor, and X, the source (or the first terminal or the like) of the transistor, the drain (or the second terminal or the like) of the transistor, and Y are provided to be connected in this order". When the connection order in a circuit configuration is defined by an expression similar to the above examples, a source (or a first terminal or the like) and a drain (or a second terminal or the like) of a transistor can be distinguished from each other to specify the technical scope. Note that these expressions are examples and there is no limitation on the expressions. Here, each of X, Y, Z1, and Z2 denotes an object (e.g., a device, an element, a circuit, a wiring, an electrode, a terminal, a conductive film, a layer, or the like).

Note that in this specification and the like, a transistor can be formed using any of a variety of substrates. The type of a substrate is not limited to a certain type. Examples of the substrate include a semiconductor substrate (e.g., a single crystal substrate or a silicon substrate), an SOI substrate, a glass substrate, a quartz substrate, a plastic substrate, a metal substrate, a stainless steel substrate, a substrate including stainless steel foil, a tungsten substrate, a substrate including tungsten foil, a flexible substrate, an attachment film, paper including a fibrous material, and a base material film. Examples of a glass substrate include a barium borosilicate glass substrate, an aluminoborosilicate glass substrate, and a soda lime glass substrate. For a flexible substrate, a flexible synthetic resin such as plastics typified by polyethylene terephthalate (PET), polyethylene naphthalate (PEN), and polyether sulfone (PES), or acrylic can be used, for example. Examples of an attachment film include attachment films formed using polypropylene, polyester, polyvinyl fluoride, polyvinyl chloride, and the like. Examples of a base film include a polyester base film, a polyamide base film, a polyimide base film, an inorganic vapor deposition film, paper, and the like. Specifically, when a transistor is formed using a semiconductor substrate, a single crystal substrate,

6

an SOI substrate, or the like, a transistor with few variations in characteristics, size, shape, or the like, high current supply capability, and a small size can be formed. By forming a circuit using such a transistor, power consumption of the circuit can be reduced or the circuit can be highly integrated.

Alternatively, a flexible substrate may be used as the substrate, and the transistor may be provided directly on the flexible substrate. Further alternatively, a separation layer may be provided between the substrate and the transistor. The separation layer can be used when part or the whole of a semiconductor device formed over the separation layer is separated from the substrate and transferred onto another substrate. In such a case, the transistor can be transferred to a substrate having low heat resistance or a flexible substrate as well. For the above separation layer, a stack including inorganic films, which are a tungsten film and a silicon oxide film, or an organic resin film of polyimide or the like formed over a substrate can be used, for example.

In other words, a transistor may be formed using one substrate, and then transferred to another substrate. Examples of a substrate to which a transistor is transferred include, in addition to the above-described substrates over which transistors can be formed, a paper substrate, a cellophane substrate, an aramid film substrate, a polyimide film substrate, a stone substrate, a wood substrate, a cloth substrate (including a natural fiber (e.g., silk, cotton, or hemp), a synthetic fiber (e.g., nylon, polyurethane, or polyester), a regenerated fiber (e.g., acetate, cupra, rayon, or regenerated polyester), or the like), a leather substrate, a rubber substrate, and the like. With the use of such a substrate, a transistor with excellent properties, a transistor with low power consumption, or a device with high durability can be formed, high heat resistance can be provided, or a reduction in weight or thinning can be achieved.

Embodiment 1

In this embodiment, a semiconductor device of one embodiment of the present invention is described with reference to drawings.

FIGS. 1A to 1C are a top view and cross-sectional views of a transistor of one embodiment of the present invention. FIG. 1A is the top view. FIG. 1B illustrates a cross section taken along dashed-dotted line A1-A2 in FIG. 1A. FIG. 1C is a cross-sectional view taken along dashed-dotted line A3-A4 in FIG. 1A. Note that for simplification of the drawing, some components are not illustrated in the top view in FIG. 1A. In some cases, the direction of the dashed-dotted line A1-A2 is referred to as a channel length direction, and the direction of the dashed-dotted line A3-A4 is referred to as a channel width direction.

A transistor 100 illustrated in FIGS. 1A to 1C and FIG. 2 includes a base insulating film 120 formed over a substrate 110; a stack in which a first oxide semiconductor layer 131 and a second oxide semiconductor layer 132 are provided in this order and which is formed over the base insulating film; a source electrode layer 140 and a drain electrode layer 150, each in contact with part of the stack; a third oxide semiconductor layer 133 which is in contact with part of each of the base insulating film 120, the stack, the source electrode layer 140, and the drain electrode layer 150; a gate insulating film 160 formed over the third oxide semiconductor layer; a gate electrode layer 170 formed over the gate insulating film; and an insulating layer 180 formed over the source electrode layer 140, the drain electrode layer 150, and the gate electrode layer 170.

Here, the first oxide semiconductor layer **131** preferably includes a crystalline layer in which c-axes are aligned in a direction perpendicular to a surface of the base insulating film **120**. The second oxide semiconductor layer **132** preferably includes a crystalline layer in which c-axes are aligned in a direction perpendicular to a top surface of the first oxide semiconductor layer **131**.

Further, the third oxide semiconductor layer **133** is formed to have a first layer in contact with the stack and a second layer over the first layer. The first layer includes a microcrystalline layer, and the second layer includes a crystalline layer in which c-axes are aligned in a direction perpendicular to a surface of the first layer.

Further, an insulating layer **185** formed using an oxide may be formed over the insulating layer **180**. The insulating layer **185** may be provided as needed and another insulating layer may be further provided thereover. The first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** are collectively referred to as an oxide semiconductor layer **130**.

Note that functions of a "source" and a "drain" of a transistor are sometimes replaced with each other when a transistor of opposite polarity is used or when the direction of current flowing is changed in circuit operation, for example. Thus, the terms "source" and "drain" can be used to denote the drain and the source, respectively, in this specification.

In addition, in the source electrode layer **140** or the drain electrode layer **150** overlapping with the oxide semiconductor layers (the first oxide semiconductor layer **131** and the second oxide semiconductor layer **132**) of the transistor of one embodiment of the present invention, the distance (ΔW) between one edge portion of the oxide semiconductor layer and one edge portion of the source electrode layer **140** or the drain electrode layer **150**, which is shown in the top view of FIG. 1A, is set shorter than or equal to 50 nm, preferably shorter than or equal to 25 nm. When ΔW is set small, oxygen contained in the base insulating film **120** can be prevented from being diffused to a metal material, which is the component of the source electrode layer **140** and the drain electrode layer **150**. Thus, unnecessary release of oxygen, in particular, excess oxygen, contained in the base insulating film **120**, can be prevented. As a result, oxygen can be efficiently supplied from the base insulating film **120** to the oxide semiconductor layer.

Then, the components of the transistor **100** of one embodiment of the present invention will be described in detail.

The substrate **110** is not limited to a simple supporting substrate, and may be a substrate where another device such as a transistor is formed. In that case, at least one of the gate electrode layer **170**, the source electrode layer **140**, and the drain electrode layer **150** of the transistor **100** may be electrically connected to the above device.

The base insulating film **120** can have a function of supplying oxygen to the oxide semiconductor layer **130** as well as a function of preventing diffusion of impurities from the substrate **110**. For this reason, the base insulating film **120** is preferably an insulating film containing oxygen and further preferably, the base insulating film **120** is an insulating film containing oxygen in which the oxygen content is higher than that in the stoichiometric composition. In the case where the substrate **110** is provided with another device as described above, the base insulating film **120** also has a function as an interlayer insulating film. In that case, the base insulating film **120** is preferably subjected to planar-

ization treatment such as chemical mechanical polishing (CMP) treatment so as to have a flat surface.

Further, in a region where a channel of the transistor **100** is formed, the oxide semiconductor layer **130** has a structure in which the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** are stacked in this order from the substrate **110** side. In addition, as illustrated in the cross-sectional view in a channel width direction in FIG. 1C, in the channel formation region, the third oxide semiconductor layer **133** is formed to cover a side surface, the top surface, and the opposite side surface of the stack including the first oxide semiconductor layer **131** and the second oxide semiconductor layer **132**. This means that, in the channel formation region, the second oxide semiconductor layer **132** is surrounded by the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133**.

Here, for the second oxide semiconductor layer **132**, for example, an oxide semiconductor whose electron affinity (an energy difference between a vacuum level and the conduction band minimum) is higher than those of the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** is used. The electron affinity can be obtained by subtracting an energy difference between the conduction band minimum and the valence band maximum (what is called an energy gap) from an energy difference between the vacuum level and the valence band maximum (what is called an ionization potential).

The first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** each contain one or more kinds of metal elements forming the second oxide semiconductor layer **132**. For example, the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** are preferably formed using an oxide semiconductor whose conduction band minimum is closer to a vacuum level than that of the second oxide semiconductor layer **132** is. Further, the energy difference of the conduction band minimum between the second oxide semiconductor layer **132** and the first oxide semiconductor layer **131** and the energy difference of the conduction band minimum between the second oxide semiconductor layer **132** and the third oxide semiconductor layer **133** are each preferably greater than or equal to 0.05 eV, 0.07 eV, 0.1 eV, or 0.15 eV and smaller than or equal to 2 eV, 1 eV, 0.5 eV, or 0.4 eV.

In such a structure, when an electric field is applied to the gate electrode layer **170**, a channel is formed in the second oxide semiconductor layer **132** whose conduction band minimum is the lowest in the oxide semiconductor layer **130**. In other words, the third oxide semiconductor layer **133** is formed between the second oxide semiconductor layer **132** and the gate insulating film **160**, whereby a structure in which the channel of the transistor is not in contact with the gate insulating film is obtained.

Further, since the first oxide semiconductor layer **131** contains one or more metal elements contained in the second oxide semiconductor layer **132**, an interface state is less likely to be formed at the interface of the second oxide semiconductor layer **132** with the first oxide semiconductor layer **131** than at the interface with the base insulating film **120** on the assumption that the second oxide semiconductor layer **132** is in contact with the base insulating film **120**. The interface state sometimes forms a channel, leading to a change in the threshold voltage of the transistor. Thus, with the first oxide semiconductor layer **131**, variations in the electrical characteristics of the transistor, such as a threshold voltage, can be reduced. Further, the reliability of the transistor can be improved.

Furthermore, since the third oxide semiconductor layer **133** contains one or more metal elements contained in the second oxide semiconductor layer **132**, scattering of carriers is less likely to occur at the interface of the second oxide semiconductor layer **132** with the third oxide semiconductor layer **133** than at the interface with the gate insulating film **160** on the assumption that the second oxide semiconductor layer **132** is in contact with the gate insulating film **160**. Thus, with the third oxide semiconductor layer **133**, the field-effect mobility of the transistor can be increased.

When each of the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** is an In-M-Zn oxide layer containing at least indium, zinc, and M (M is a metal such as Al, Ti, Ga, Ge, Y, Zr, Sn, La, Ce, or Hf), the atomic ratio of M to In or Zn in the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** is preferably higher than that in the second oxide semiconductor layer **132**. Specifically, the atomic ratio of M to In or Zn in the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** is 1.5 times or more, preferably 2 times or more, further preferably 3 times or more as much as that in the second oxide semiconductor layer **132**. The metal M is more strongly bonded to oxygen than In or Zn is and thus has a function of suppressing generation of an oxygen vacancy in an oxide semiconductor layer. That is, an oxygen vacancy is less likely to be generated in the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** than in the second oxide semiconductor layer **132**.

Note that when each of the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** is an In-M-Zn oxide layer containing at least indium, zinc, and M (M is a metal such as Al, Ti, Ga, Ge, Y, Zr, Sn, La, Ce, or Hf), and the first oxide semiconductor layer **131** has an atomic ratio of In to M and Zn which is $x_1:y_1:z_1$, the second oxide semiconductor layer **132** has an atomic ratio of In to M and Zn which is $x_2:y_2:z_2$, and the third oxide semiconductor layer **133** has an atomic ratio of In to M and Zn which is $x_3:y_3:z_3$, each of y_1/x_1 and y_3/x_3 is preferably larger than y_2/x_2 . Each of y_1/x_1 and y_3/x_3 is 1.5 times or more, preferably 2 times or more, further preferably 3 times or more as large as y_2/x_2 . At this time, when y_2 is greater than or equal to x_2 in the second oxide semiconductor layer **132**, the transistor can have stable electrical characteristics. However, when y_2 is 3 times or more as large as x_2 , the field-effect mobility of the transistor is reduced; accordingly, y_2 is preferably less than 3 times x_2 .

Note that in this specification, an atomic ratio used for describing the composition of an oxide semiconductor layer can be also used as the atomic ratio of a base material. In the case where an oxide semiconductor layer is deposited by a sputtering method using an oxide semiconductor material as a target, the composition of the oxide semiconductor layer might be different from that of the target, which is a base material, depending on the kind or a ratio of a sputtering gas, the density of the target, or deposition conditions. Thus, in this specification, an atomic ratio used for describing the composition of an oxide semiconductor layer is also used as the atomic ratio of a base material. For example, in the case where a sputtering method is used for deposition, an In—Ga—Zn oxide film whose atomic ratio of In to Ga and Zn is 1:1:1 can be also understood as an In—Ga—Zn oxide film formed using an In—Ga—Zn oxide material whose atomic ratio of In to Ga and Zn is 1:1:1 as a target.

Further, in the case where Zn and O are not taken into consideration, the proportion of In and the proportion of M in each of the first oxide semiconductor layer **131** and the

third oxide semiconductor layer **133** are preferably less than 50 atomic % and greater than or equal to 50 atomic %, respectively, and further preferably less than 25 atomic % and greater than or equal to 75 atomic %, respectively. In addition, in the case where Zn and O are not taken into consideration, the proportion of In and the proportion of M in the second oxide semiconductor layer **132** are preferably greater than or equal to 25 atomic % and less than 75 atomic %, respectively, and further preferably greater than or equal to 34 atomic % and less than 66 atomic %, respectively.

The thicknesses of the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** are each greater than or equal to 1 nm and less than or equal to 100 nm, preferably greater than or equal to 3 nm and less than or equal to 50 nm. The thickness of the second oxide semiconductor layer **132** is greater than or equal to 1 nm and less than or equal to 200 nm, preferably greater than or equal to 3 nm and less than or equal to 100 nm, further preferably greater than or equal to 3 nm and less than or equal to 50 nm.

For the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133**, an oxide semiconductor containing indium, zinc, and gallium can be used, for example. Note that the second oxide semiconductor layer **132** preferably contains indium because carrier mobility can be increased.

Accordingly, with the oxide semiconductor layer **130** having a stacked-layer structure including the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133**, a channel can be formed in the second oxide semiconductor layer **132**; thus, the transistor can have a high field-effect mobility and stable electrical characteristics.

In a band structure, the conduction band minimums of the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** are continuous. This can be understood also from the fact that the compositions of the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** are close to one another and oxygen is easily diffused among the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133**. Thus, the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** have a continuous physical property although they have different compositions and form a stack. In the drawings, interfaces between the oxide semiconductor layers of the stack are indicated by dotted lines.

The oxide semiconductor layer **130** in which layers containing the same main components are stacked is formed to have not only a simple stacked-layer structure of the layers but also a continuous energy band (here, in particular, a well structure having a U shape in which the conduction band minimums are continuous). In other words, the stacked-layer structure is formed such that there exists no impurity that forms a defect level such as a trap center or a recombination center at each interface. If impurities exist between the stacked oxide semiconductor layers, the continuity of the energy band is lost and carriers disappear by a trap or recombination at the interface.

An In—Ga—Zn oxide whose atomic ratio of In to Ga and Zn is 1:3:2, 1:3:3, 1:3:4, 1:3:6, 1:6:4, or 1:9:6 can be used for the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** and an In—Ga—Zn oxide whose atomic ratio of In to Ga and Zn is 1:1:1, 5:5:6, 3:1:2, or the like can be used for the second oxide semiconductor layer **132**, for example.

The second oxide semiconductor layer 132 of the oxide semiconductor layer 130 serves as a well, so that a channel is formed in the second oxide semiconductor layer 132 in a transistor including the oxide semiconductor layer 130. Note that since the conduction band minimums are continuous, the oxide semiconductor layer 130 can also be referred to as a U-shaped well. Further, a channel formed to have such a structure can also be referred to as a buried channel.

Note that trap levels due to impurities or defects might be formed in the vicinity of the interface between an insulating film such as a silicon oxide film and each of the first oxide semiconductor layer 131 and the third oxide semiconductor layer 133. The second oxide semiconductor layer 132 can be distanced away from the trap levels owing to existence of the first oxide semiconductor layer 131 and the third oxide semiconductor layer 133.

However, when the energy differences between the conduction band minimum of the second oxide semiconductor layer 132 and the conduction band minimum of each of the first oxide semiconductor layer 131 and the third oxide semiconductor layer 133 are small, an electron in the second oxide semiconductor layer 132 might reach the trap level by passing over the energy differences. When the electron is trapped in the trap level, a negative fixed charge is generated at the interface with the insulating film, whereby the threshold voltage of the transistor is shifted in the positive direction.

Thus, to reduce fluctuations in the threshold voltage of the transistor, energy differences of at least certain values between the conduction band minimum of the second oxide semiconductor layer 132 and the conduction band minimum of each of the first oxide semiconductor layer 131 and the third oxide semiconductor layer 133 are necessary. Each of the energy differences is preferably greater than or equal to 0.1 eV, further preferably greater than or equal to 0.15 eV.

Note that each of the first oxide semiconductor layer 131, the second oxide semiconductor layer 132, and the third oxide semiconductor layer 133 preferably includes a crystalline layer in which c-axes are aligned. A film containing the crystalline layer can provide a transistor with stable electrical characteristics.

In the case where an In—Ga—Zn oxide is used for the oxide semiconductor layer 130, it is preferable that the third oxide semiconductor layer 133 contain less In than the second oxide semiconductor layer 132 so that diffusion of In to the gate insulating film is prevented.

The above-described buried channel is formed in the transistor of one embodiment of the present invention. In addition, as in the transistor illustrated in FIG. 2, the third oxide semiconductor layer 133 includes a microcrystalline layer 133a in contact with the base insulating film 120 and the stack including the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132, and a crystalline layer 133b in which c-axes are aligned in a direction perpendicular to a surface of the microcrystalline layer.

FIG. 3 illustrates the details of the band structure of the oxide semiconductor layers (in the B1-B2 direction in FIG. 2) having such a structure. Here, Evac represents energy of the vacuum level, Ec11 and Ec12 each represent the conduction band minimum of the silicon oxide film, EcS1 represents the conduction band minimum of the first oxide semiconductor layer 131, EcS2 represents the conduction band minimum of the second oxide semiconductor layer 132, and EcS3 represents the conduction band minimum of the third oxide semiconductor layer 133.

Energy does not change suddenly between EcS1 and EcS2 and between EcS3 and EcS2, and gradually starts and stops changing.

This is because the constituents of the oxide semiconductor layers are diffused interactively between the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132 and between the third oxide semiconductor layer 133 and the second oxide semiconductor layer 132, which leads to formation of a region whose composition is intermediate between the compositions of the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132 or a region whose composition is intermediate between the compositions of the third oxide semiconductor layer 133 and the second oxide semiconductor layer 132.

Thus, as illustrated in FIG. 3, a channel formed in the second oxide semiconductor layer 132 is formed in a region 132b which is positioned at an inner side than the interface between the third oxide semiconductor layer 133 and the second oxide semiconductor layer 132 and the interface between the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132. With such a structure, a carrier can be prevented from being trapped or recombined even when a defect or an impurity exists at either one of the interfaces.

In the third oxide semiconductor layer 133, a region in contact with a stack including the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132 includes the microcrystalline layer 133a. The density of the microcrystalline layer is lower than that of the crystalline layer 133b, which is formed over the microcrystalline layer; thus, the constituents of the second oxide semiconductor layer 132 are easily diffused to the third oxide semiconductor layer 133 side. As a result, the region whose composition is intermediate between the compositions of the third oxide semiconductor layer 133 and the second oxide semiconductor layer 132 becomes large. Thus, the channel formed in the second oxide semiconductor layer 132 is positioned further apart from the interface between the third oxide semiconductor layer 133 and the second oxide semiconductor layer 132 toward the center of the second oxide semiconductor layer 132, and a malfunction which occurs when a defect or an impurity exists at the interface can be avoided more effectively.

In the case where the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132 each include a crystalline layer in which c-axes are aligned, oxygen is relatively likely to be diffused since the density of the microcrystalline layer 133a is lower than that of the crystalline layer. Accordingly, oxygen can be efficiently supplied from the base insulating film 120 to the second oxide semiconductor layer 132 to be a channel with the use of the microcrystalline layer 133a as a path, and an oxygen vacancy can be filled with oxygen.

Further, in the crystalline layer 133b in the third oxide semiconductor layer 133, c-axes are aligned in the direction perpendicular to the surface of the microcrystalline layer 133a. Thus, when the second oxide semiconductor layer 132 is formed to have a curved surface, a channel region in the second oxide semiconductor layer 132 can be densely covered by crystals whose c-axes are aligned.

FIG. 4A is a cross-sectional view in the channel width direction of the transistor, which schematically illustrates part of a crystal structure of a stack including the second oxide semiconductor layer 132 formed to have a curved surface, the microcrystalline layer 133a covering the second oxide semiconductor layer, and the crystalline layer 133b formed over the microcrystalline layer. Here, the second

13

oxide semiconductor layer **132** is a crystalline layer in which c-axes are aligned in a direction perpendicular to a surface of the first oxide semiconductor layer **131** (not illustrated).

When the second oxide semiconductor layer **132** is formed to have a curved surface as illustrated in FIG. 4A, the third oxide semiconductor layer **133** can be formed to have the dense crystalline layer **133b** in which c-axes are aligned in the direction perpendicular to the curved surface, with the microcrystalline layer **133a** is provided between the second oxide semiconductor layer **132** and the dense crystalline layer **133b**. Such a structure can improve an effect of suppressing release of oxygen from the second oxide semiconductor layer **132** or an effect of confining oxygen released from the base insulating film **120** by the third oxide semiconductor layer **133**; thus, an oxygen vacancy in the second oxide semiconductor layer **132** can be efficiently filled with oxygen.

Note that in the case where the second oxide semiconductor layer **132** is formed not to have a curved surface as illustrated in FIG. 4B, a region **233** in which crystals are sparse is formed at an intersection of the crystalline layer **133b** formed over the top surface of the second oxide semiconductor layer **132** and the crystalline layer **133b** that is formed to face a side surface of the second oxide semiconductor layer **132**, in the third oxide semiconductor layer **133**. Thus, oxygen contained in the second oxide semiconductor layer **132** and oxygen supplied from the base insulating film **120** to the second oxide semiconductor layer **132** are likely to be released through the region **233**, in which case an oxygen vacancy in the second oxide semiconductor layer **132** cannot be efficiently filled with oxygen.

Note that stable electrical characteristics can be effectively imparted to a transistor in which an oxide semiconductor layer serves as a channel by reducing the concentration of impurities in the oxide semiconductor layer to make the oxide semiconductor layer intrinsic or substantially intrinsic. The term "substantially intrinsic" refers to the state where an oxide semiconductor layer has a carrier density lower than $1 \times 10^{17}/\text{cm}^3$, preferably lower than $1 \times 10^{15}/\text{cm}^3$, further preferably lower than $1 \times 10^{13}/\text{cm}^3$.

Further, in the oxide semiconductor layer, hydrogen, nitrogen, carbon, silicon, and a metal element other than main components are impurities. For example, hydrogen and nitrogen form donor levels to increase the carrier density, and silicon forms impurity levels in the oxide semiconductor layer. The impurity levels serve as traps and might cause the electrical characteristics of the transistor to deteriorate. Thus, it is preferable to reduce the concentration of the impurities in the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133**, and at interfaces between the layers.

In order to make the oxide semiconductor layer intrinsic or substantially intrinsic, in SIMS (secondary ion mass spectrometry), for example, the concentration of silicon at a certain depth of the oxide semiconductor layer or in a region of the oxide semiconductor layer is preferably lower than 1×10^{19} atoms/ cm^3 , further preferably lower than 5×10^{18} atoms/ cm^3 , still further preferably lower than 1×10^{18} atoms/ cm^3 . Further, the concentration of hydrogen at a certain depth of the oxide semiconductor layer or in a region of the oxide semiconductor layer is preferably lower than or equal to 2×10^{20} atoms/ cm^3 , further preferably lower than or equal to 5×10^{19} atoms/ cm^3 , still further preferably lower than or equal to 1×10^{19} atoms/ cm^3 , yet still further preferably lower than or equal to 5×10^{18} atoms/ cm^3 . Further, the concentration of nitrogen at a certain depth of the oxide semiconductor

14

layer or in a region of the oxide semiconductor layer is preferably lower than 5×10^{19} atoms/ cm^3 , further preferably lower than or equal to 5×10^{18} atoms/ cm^3 , still further preferably lower than or equal to 1×10^{18} atoms/ cm^3 , yet still further preferably lower than or equal to 5×10^{17} atoms/ cm^3 .

In the case where the oxide semiconductor layer includes crystals, high concentration of silicon or carbon might reduce the crystallinity of the oxide semiconductor layer. In order not to reduce the crystallinity of the oxide semiconductor layer, for example, the concentration of silicon at a certain depth of the oxide semiconductor layer or in a region of the oxide semiconductor layer may be lower than 1×10^{19} atoms/ cm^3 , preferably lower than 5×10^{18} atoms/ cm^3 , further preferably lower than 1×10^{18} atoms/ cm^3 . Further, the concentration of carbon at a certain depth of the oxide semiconductor layer or in a region of the oxide semiconductor layer may be lower than 1×10^{19} atoms/ cm^3 , preferably lower than 5×10^{18} atoms/ cm^3 , further preferably lower than 1×10^{18} atoms/ cm^3 , for example.

A transistor in which the above-described highly purified oxide semiconductor layer is used for a channel formation region has an extremely low off-state current. In the case where the voltage between a source and a drain is set to approximately 0.1 V, 5 V, or 10 V, for example, the off-state current standardized on the channel width of the transistor can be as low as several yoctoamperes per micrometer to several zeptoamperes per micrometer.

Note that as the gate insulating film of the transistor, an insulating film containing silicon is used in many cases; thus, it is preferable that, as in the transistor of one embodiment of the present invention, a region of the oxide semiconductor layer, which serves as a channel, be not in contact with the gate insulating film for the above-described reason. In the case where a channel is formed at the interface between the gate insulating film and the oxide semiconductor layer, scattering of carriers occurs at the interface, whereby the field-effect mobility of the transistor is reduced in some cases. Also from the view of the above, it is preferable that the region of the oxide semiconductor layer, which serves as a channel, be separated from the gate insulating film.

For the source electrode layer **140** and the drain electrode layer **150**, a conductive material which is easily bonded to oxygen is preferably used. For example, Al, Cr, Cu, Ta, Ti, Mo, or W can be used. Among the materials, in particular, it is preferable to use Ti which is easily bonded to oxygen or to use W with a high melting point, which allows subsequent process temperatures to be relatively high. Note that the conductive material which is easily bonded to oxygen includes, in its category, a material to which oxygen is easily diffused.

When the conductive material which is easily bonded to oxygen is in contact with an oxide semiconductor layer, a phenomenon occurs in which oxygen in the oxide semiconductor layer is diffused to the conductive material which is easily bonded to oxygen. The phenomenon noticeably occurs when the temperature is high. Since the manufacturing process of the transistor involves a heat treatment step, the above phenomenon causes generation of oxygen vacancies in the vicinity of a region which is in the oxide semiconductor layer and is in contact with the source electrode layer or the drain electrode layer. The oxygen vacancies bond to hydrogen slightly contained in the layer, whereby the region is changed to an n-type region. Thus, the n-type region can serve as a source or a drain of the transistor.

15

The n-type region is illustrated in an enlarged cross-sectional view of the transistor (showing part of a cross section in the channel length direction, which is near the source electrode layer 140) in FIG. 5. A boundary 135 indicated by a dotted line in the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132 is a boundary between an intrinsic semiconductor region and an n-type semiconductor region. In the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132, a region near the source electrode layer 140 becomes an n-type region. The boundary 135 is schematically illustrated here, but actually, the boundary is not clearly seen in some cases. Although FIG. 5 shows that part of the boundary 135 extends in the lateral direction in the second oxide semiconductor layer 132, a region in the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132, which is sandwiched between the source electrode layer 140 and the base insulating film 120, becomes n-type entirely in the thickness direction, in some cases.

In the case of forming a transistor with an extremely short channel length, an n-type region which is formed by the generation of oxygen vacancies might extend in the channel length direction of the transistor. In that case, the electrical characteristics of the transistor change; for example, the threshold voltage is shifted, or on and off states of the transistor cannot be controlled with the gate voltage (in which case the transistor is turned on). Accordingly, when a transistor with an extremely short channel length is formed, it is not always preferable that a conductive material easily bonded to oxygen be used for a source electrode layer and a drain electrode layer.

In such a case, a conductive material which is less likely to be bonded to oxygen than the above material can be used for the source electrode layer 140 and the drain electrode layer 150. As the conductive material which is not easily bonded to oxygen, for example, a material containing tantalum nitride, titanium nitride, gold, platinum, palladium, or ruthenium or the like can be used. Note that in the case where the conductive material is in contact with the second oxide semiconductor layer 132, the source electrode layer 140 and the drain electrode layer 150 may each have a structure in which the conductive material which is not easily bonded to oxygen and the above-described conductive material that is easily bonded to oxygen are stacked.

The gate insulating film 160 can be formed using an insulating film containing one or more of aluminum oxide, magnesium oxide, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, gallium oxide, germanium oxide, yttrium oxide, zirconium oxide, lanthanum oxide, neodymium oxide, hafnium oxide, and tantalum oxide. The gate insulating film 160 may be a stack including any of the above materials.

For the gate electrode layer 170, a conductive film formed using Al, Ti, Cr, Co, Ni, Cu, Y, Zr, Mo, Ru, Ag, Ta, W, or the like can be used. The gate electrode layer may be a stack including any of the above materials. Alternatively, a conductive film containing nitrogen may be used for the gate electrode layer.

The insulating layer 180 is preferably formed over the gate insulating film 160 and the gate electrode layer 170. The insulating layer is preferably formed using aluminum oxide. The aluminum oxide film has a high blocking effect of preventing penetration of both oxygen and impurities such as hydrogen and moisture. Accordingly, during and after the manufacturing process of the transistor, the aluminum oxide film can suitably function as a protective film that has effects of preventing entry of impurities such as hydrogen and

16

moisture, which cause variations in the electrical characteristics of the transistor, into the oxide semiconductor layer 130, preventing release of oxygen, which is a main component of the oxide semiconductor layer 130, from the oxide semiconductor layer, and preventing unnecessary release of oxygen from the base insulating film 120. Further, oxygen contained in the aluminum oxide film can be diffused in the oxide semiconductor layer.

Further, the insulating layer 185 is preferably formed over the insulating layer 180. The insulating layer 185 can be formed using an insulating film containing one or more of magnesium oxide, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, gallium oxide, germanium oxide, yttrium oxide, zirconium oxide, lanthanum oxide, neodymium oxide, hafnium oxide, and tantalum oxide. The insulating layer 185 may be a stack including any of the above materials.

Here, the insulating layer 185 preferably contains excess oxygen. An insulating layer containing excess oxygen refers to an insulating layer from which oxygen can be released by heat treatment or the like. The insulating layer containing excess oxygen is, for example, a film in which the amount of released oxygen when converted into oxygen atoms is 1.0×10^{19} atoms/cm³ or more in thermal desorption spectroscopy analysis. In the thermal desorption spectroscopy analysis, heat treatment is performed at a surface temperature of higher than or equal to 100° C. and lower than or equal to 700° C., preferably higher than or equal to 100° C. and lower than or equal to 500° C. Oxygen released from the insulating layer can be diffused to the channel formation region in the oxide semiconductor layer 130 through the gate insulating film 160, so that oxygen vacancies formed in the channel formation region can be filled with the oxygen. In this manner, the electrical characteristics of the transistor can be stable.

High integration of a semiconductor device requires miniaturization of a transistor. However, it is known that miniaturization of a transistor causes deterioration of the electrical characteristics of the transistor. In particular, a reduction in on-state current, which is directly caused by a decrease in channel width, is significant.

However, in the transistor of one embodiment of the present invention, as described above, the third oxide semiconductor layer 133 is formed so as to cover a region where a channel is formed in the second oxide semiconductor layer 132, and the channel formation layer and the gate insulating film are not in contact with each other. Accordingly, scattering of carriers at the interface between the channel formation layer and the gate insulating film can be reduced and the field-effect mobility of the transistor can be increased.

In addition, the electrical characteristics of the transistor of one embodiment of the present invention can be particularly improved with a structure as illustrated in a cross-sectional view in the channel width direction in FIG. 2, in which the length of the top surface (W_T) of the second oxide semiconductor layer 132 in the channel width direction is as small as its thickness.

In the case where W_T is small as in a transistor illustrated in FIG. 2, for example, an electric field from the gate electrode layer 170 to the side surface of the second oxide semiconductor layer 132 is applied to the entire second oxide semiconductor layer 132; thus, a channel is formed equally in the side and top surfaces of the second oxide semiconductor layer 132.

In the case of a transistor in which W_T is small, the channel width can be defined as the sum of W_T and the

17

lengths of the side surfaces (W_{S1} and W_{S2}) of the second oxide semiconductor layer 132 in the channel width direction (i.e., $W_T + W_{S1} + W_{S2}$), and on-state current flows in the transistor in accordance with the channel width. In the case where W_T is extremely small, current flows in the entire second oxide semiconductor layer 132.

That is, the transistor of one embodiment of the present invention in which W_T is small can have higher on-state current than the conventional transistor owing to both of an effect of suppressing scattering of carriers and an effect of extending the channel width.

Note that in order to efficiently increase the on-state current of the transistor when W_{S1} and W_{S2} are represented by W_S ($W_{S1} = W_{S2} = W_S$), a relation $0.3W_S \leq W_T \leq 3W_S$ (W_T is greater than or equal to $0.3W_S$ and less than or equal to $3W_S$) is satisfied. Further, W_T/W_S is preferably greater than or equal to 0.5 and less than or equal to 1.5, further preferably greater than or equal to 0.7 and less than or equal to 1.3. In the case where $W_T/W_S > 3$, the S value and the off-state current might be increased.

As described above, with the transistor of one embodiment of the present invention, sufficiently high on-state current can be obtained even when the transistor is miniaturized.

In the transistor of one embodiment of the present invention, the second oxide semiconductor layer 132 is formed over the first oxide semiconductor layer 131, so that an interface state is less likely to be formed. In addition, impurities do not enter the second oxide semiconductor layer 132 from above and below because the second oxide semiconductor layer 132 is an intermediate layer in a three-layer structure. Since the second oxide semiconductor layer 132 is surrounded by the first oxide semiconductor layer 131 and the third oxide semiconductor layer 133, not only the on-state current of the transistor can be increased but the threshold voltage can be stabilized and the S value can be reduced. Thus, I_{cut} (current when gate voltage VG is 0 V) can be reduced and power consumption of the semiconductor device can be reduced. Further, the threshold voltage of the transistor becomes stable; thus, long-term reliability of the semiconductor device can be improved.

The transistor of one embodiment of the present invention may include a conductive film 172 between the oxide semiconductor layer 130 and the substrate 110 as illustrated in FIG. 6. When the conductive film is used as a second gate electrode, the on-state current can be further increased and the threshold voltage can be controlled. In order to increase the on-state current, for example, the gate electrode layer 170 and the conductive film 172 are set to have the same potential, and the transistor is driven as a dual-gate transistor. Further, to control the threshold voltage, a fixed potential, which is different from a potential of the gate electrode layer 170, is supplied to the conductive film 172.

This embodiment can be combined as appropriate with any of the other embodiments and an example in this specification.

Embodiment 2

In this embodiment, a method for forming the transistor 100, which is described in Embodiment 1 with reference to FIGS. 1A to 1C, is described with reference to FIGS. 7A to 7C and FIGS. 8A to 8C.

For the substrate 110, a glass substrate, a ceramic substrate, a quartz substrate, a sapphire substrate, or the like can be used. Alternatively, a single crystal semiconductor substrate or a polycrystalline semiconductor substrate made of

18

silicon, silicon carbide, or the like, a compound semiconductor substrate made of silicon germanium or the like, a silicon-on-insulator (SOI) substrate, or the like can be used. Further alternatively, any of these substrates further provided with a semiconductor element can be used.

The base insulating film 120 can be formed by a plasma CVD method, a sputtering method, or the like using an oxide insulating film of aluminum oxide, magnesium oxide, silicon oxide, silicon oxynitride, gallium oxide, germanium oxide, yttrium oxide, zirconium oxide, lanthanum oxide, neodymium oxide, hafnium oxide, tantalum oxide, or the like; a nitride insulating film of silicon nitride, silicon nitride oxide, aluminum nitride, aluminum nitride oxide, or the like; or a film in which any of the above materials are mixed. Alternatively, a stack including any of the above materials may be used, and at least an upper layer of the base insulating film 120 which is in contact with the oxide semiconductor layer 130 is preferably formed using a material containing excess oxygen that might serve as a supply source of oxygen to the oxide semiconductor layer 130.

Oxygen may be added to the base insulating film 120 by an ion implantation method, an ion doping method, a plasma immersion ion implantation method, or the like. Adding oxygen enables the base insulating film 120 to supply oxygen much easily to the oxide semiconductor layer 130.

In the case where a surface of the substrate 110 is made of an insulator and there is no influence of impurity diffusion to the oxide semiconductor layer 130 to be formed later, the base insulating film 120 is not necessarily provided.

Next, a first oxide semiconductor film 331 to be the first oxide semiconductor layer 131 and a second oxide semiconductor film 332 to be the second oxide semiconductor layer 132 are deposited over the base insulating film 120 by a sputtering method, a CVD method, an MBE method, an ALD method, or a PLD method (see FIG. 7A).

Subsequently, the first oxide semiconductor film 331 and the second oxide semiconductor film 332 are selectively etched to form the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132 (see FIG. 7B). At this time, the base insulating film 120 may also be etched slightly as illustrated in FIG. 7B. The slightly etched base insulating film 120 enables the second oxide semiconductor layer 132 to be easily covered by the gate electrode that is formed later. Further, the second oxide semiconductor layer 132 is formed to have a curvature from its top surface to its side surface in the cross section in the channel width direction of the transistor.

Note that when the first oxide semiconductor film 331 and the second oxide semiconductor film 332 are selectively etched, not only a photoresist but also a hard mask such as a metal film can be used. In addition, an organic resin may be formed over the metal film. As the metal film, for example, a tungsten film with a thickness of approximately 5 nm can be used.

For the etching, dry etching in which a difference between the etching rate of the first oxide semiconductor film 331 and that of the second oxide semiconductor film 332 is small is preferably used.

In order to form a continuous energy band in a stack including the first oxide semiconductor layer 131 and the second oxide semiconductor layer 132, the layers are preferably formed successively without exposure to the air with the use of a multi-chamber deposition apparatus (e.g., a sputtering apparatus) including a load lock chamber. It is preferable that each chamber of the sputtering apparatus be able to be evacuated to a high vacuum (to approximately higher than or equal to 5×10^{-7} Pa and lower than or equal to

1×10^{-4} Pa) by an adsorption vacuum pump such as a cryopump and that the chamber be able to heat a substrate over which a film is to be deposited to 100°C . or higher, preferably 500°C . or higher, so that water and the like acting as impurities of an oxide semiconductor are removed as much as possible. Alternatively, a combination of a turbo molecular pump and a cold trap is preferably used to prevent back-flow of a gas containing a carbon component, moisture, or the like from an exhaust system into the chamber.

Not only high vacuum evacuation of the chamber but also high purity of a sputtering gas is necessary to obtain a highly purified intrinsic oxide semiconductor. An oxygen gas or an argon gas used as the sputtering gas is highly purified to have a dew point of -40°C . or lower, preferably -80°C . or lower, further preferably -100°C . or lower, so that entry of moisture and the like into the oxide semiconductor layer can be prevented as much as possible.

For the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** formed in a later step, any of the materials described in Embodiment 1 can be used. For example, an In—Ga—Zn oxide whose atomic ratio of In to Ga and Zn is 1:3:6, 1:3:4, 1:3:3, or 1:3:2 can be used for the first oxide semiconductor layer **131**, an In—Ga—Zn oxide whose atomic ratio of In to Ga and Zn is 1:1:1 or 5:5:6 can be used for the second oxide semiconductor layer **132**, and an In—Ga—Zn oxide whose atomic ratio of In to Ga and Zn is 1:3:6, 1:3:4, 1:3:3, or 1:3:2 can be used for the third oxide semiconductor layer **133**.

An oxide semiconductor that can be used for each of the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** preferably contains at least indium (In) or zinc (Zn). Alternatively, the oxide semiconductor preferably contains both In and Zn. In order to reduce variations in the electrical characteristics of the transistor including the oxide semiconductor, the oxide semiconductor preferably contains a stabilizer in addition to In and/or Zn.

Examples of a stabilizer include gallium (Ga), tin (Sn), hafnium (Hf), aluminum (Al), and zirconium (Zr). Other examples of a stabilizer are lanthanoids such as lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), and lutetium (Lu).

As the oxide semiconductor, for example, any of the following can be used: indium oxide, tin oxide, zinc oxide, an In—Zn oxide, a Sn—Zn oxide, an Al—Zn oxide, a Zn—Mg oxide, a Sn—Mg oxide, an In—Mg oxide, an In—Ga oxide, an In—Ga—Zn oxide, an In—Al—Zn oxide, an In—Sn—Zn oxide, a Sn—Ga—Zn oxide, an Al—Ga—Zn oxide, a Sn—Al—Zn oxide, an In—Hf—Zn oxide, an In—La—Zn oxide, an In—Ce—Zn oxide, an In—Pr—Zn oxide, an In—Nd—Zn oxide, an In—Sm—Zn oxide, an In—Eu—Zn oxide, an In—Gd—Zn oxide, an In—Tb—Zn oxide, an In—Dy—Zn oxide, an In—Ho—Zn oxide, an In—Er—Zn oxide, an In—Tm—Zn oxide, an In—Yb—Zn oxide, an In—Lu—Zn oxide, an In—Sn—Ga—Zn oxide, an In—Hf—Ga—Zn oxide, an In—Al—Ga—Zn oxide, an In—Sn—Al—Zn oxide, an In—Sn—Hf—Zn oxide, and an In—Hf—Al—Zn oxide.

Note that here, for example, an “In—Ga—Zn oxide” means an oxide containing In, Ga, and Zn as its main components. The In—Ga—Zn oxide may contain a metal element other than In, Ga, and Zn. Further, in this specification, a film formed using an In—Ga—Zn oxide is also referred to as an IGZO film.

Alternatively, a material represented by $\text{InMO}_3(\text{ZnO})_m$ ($m > 0$, where m is not an integer) may be used. Note that M represents one or more metal elements selected from Ga, Y, Zr, La, Ce, and Nd. Further alternatively, a material represented by $\text{In}_2\text{SnO}_5(\text{ZnO})_n$ ($n > 0$, where n is an integer) may be used.

Note that as described in Embodiment 1 in detail, materials are selected so that the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** each have an electron affinity lower than that of the second oxide semiconductor layer **132**.

The oxide semiconductor layers are each preferably formed by a sputtering method. As a sputtering method, an RF sputtering method, a DC sputtering method, an AC sputtering method, or the like can be used.

In the case of using an In—Ga—Zn oxide, a material whose atomic ratio of In to Ga and Zn is any of 1:1:1, 2:2:1, 3:1:2, 5:5:6, 1:3:2, 1:3:3, 1:3:4, 1:3:6, 1:4:3, 1:5:4, 1:6:6, 2:1:3, 1:6:4, 1:9:6, 1:1:4, and 1:1:2 is used for the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and/or the third oxide semiconductor layer **133** so that the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** each have an electron affinity lower than that of the second oxide semiconductor layer **132**.

Note that for example, in the case where the composition of an oxide containing In, Ga, and Zn at the atomic ratio, $\text{In:Ga:Zn} = a:b:c$ ($a+b+c=1$), is in the neighborhood of the composition of an oxide containing In, Ga, and Zn at the atomic ratio, $\text{In:Ga:Zn} = A:B:C$ ($A+B+C=1$), a , b , and c satisfy the following relation: $(a-A)^2 + (b-B)^2 + (c-C)^2 \leq r^2$, and r may be 0.05, for example. The same applies to other oxides.

The indium content of the second oxide semiconductor layer **132** is preferably higher than the indium content of the first oxide semiconductor layer **131** and the indium content of the third oxide semiconductor layer **133**. In an oxide semiconductor, the s orbital of heavy metal mainly contributes to carrier transfer, and when the proportion of In in the oxide semiconductor is increased, overlap of the s orbitals is likely to be increased. Thus, an oxide having a composition in which the proportion of In is higher than that of Ga has higher mobility than an oxide having a composition in which the proportion of In is equal to or lower than that of Ga. For this reason, with the use of an oxide having a high indium content for the second oxide semiconductor layer **132**, a transistor having high mobility can be achieved.

A structure of an oxide semiconductor film is described below.

Note that in this specification, a term “parallel” indicates that the angle formed between two straight lines is greater than or equal to -10° and less than or equal to 10° , and accordingly also includes the case where the angle is greater than or equal to -5° and less than or equal to 5° . In addition, a term “perpendicular” indicates that the angle formed between two straight lines is greater than or equal to 80° and less than or equal to 100° , and accordingly includes the case where the angle is greater than or equal to 85° and less than or equal to 95° .

In this specification, the trigonal and rhombohedral crystal systems are included in the hexagonal crystal system.

An oxide semiconductor film is classified roughly into a single-crystal oxide semiconductor film and a non-single-crystal oxide semiconductor film. The non-single-crystal oxide semiconductor film includes any of a c -axis aligned crystalline oxide semiconductor (CAAC-OS) film, a

polycrystalline oxide semiconductor film, a microcrystalline oxide semiconductor film, an amorphous oxide semiconductor film, and the like.

First, a CAAC-OS film is described.

The CAAC-OS film is one of oxide semiconductor films including a plurality of crystal parts, and most of the crystal parts each fit inside a cube whose one side is less than 100 nm. Thus, there is a case where a crystal part included in the CAAC-OS film fits inside a cube whose one side is less than 10 nm, less than 5 nm, or less than 3 nm.

In a transmission electron microscope (TEM) image of the CAAC-OS film, a boundary between crystal parts, that is, a grain boundary is not clearly observed. Thus, in the CAAC-OS film, a reduction in electron mobility due to the grain boundary is less likely to occur.

According to the TEM image of the CAAC-OS film observed in a direction substantially parallel to a sample surface (cross-sectional TEM image), metal atoms are arranged in a layered manner in the crystal parts. Each metal atom layer has a morphology reflected by a surface over which the CAAC-OS film is formed (hereinafter, a surface over which the CAAC-OS film is formed is referred to as a formation surface) or a top surface of the CAAC-OS film, and is arranged in parallel to the formation surface or the top surface of the CAAC-OS film.

On the other hand, according to the TEM image of the CAAC-OS film observed in a direction substantially perpendicular to the sample surface (plan TEM image), metal atoms are arranged in a triangular or hexagonal configuration in the crystal parts. However, there is no regularity of arrangement of metal atoms between different crystal parts.

From the results of the cross-sectional TEM image and the plan TEM image, alignment is found in the crystal parts in the CAAC-OS film.

A CAAC-OS film is subjected to structural analysis with an X-ray diffraction (XRD) apparatus. For example, when the CAAC-OS film including an InGaZnO₄ crystal is analyzed by an out-of-plane method, a peak appears frequently when the diffraction angle (2θ) is around 31°. This peak is derived from the (009) plane of the InGaZnO₄ crystal, which indicates that crystals in the CAAC-OS film have c-axis alignment, and that the c-axes are aligned in a direction substantially perpendicular to the formation surface or the top surface of the CAAC-OS film.

On the other hand, when the CAAC-OS film is analyzed by an in-plane method in which an X-ray enters a sample in a direction substantially perpendicular to the c-axis, a peak appears frequently when 2θ is around 56°. This peak is derived from the (110) plane of the InGaZnO₄ crystal. Here, analysis (φ scan) is performed under conditions where the sample is rotated around a normal vector of a sample surface as an axis (φ axis) with 2θ fixed at around 56°. In the case where the sample is a single-crystal oxide semiconductor film of InGaZnO₄, six peaks appear. The six peaks are derived from crystal planes equivalent to the (110) plane. On the other hand, in the case of a CAAC-OS film, a peak is not clearly observed even when (p scan) is performed with 2θ fixed at around 56°.

According to the above results, in the CAAC-OS film having c-axis alignment, while the directions of a-axes and b-axes are different between crystal parts, the c-axes are aligned in a direction parallel to a normal vector of a formation surface or a normal vector of a top surface. Thus, each metal atom layer arranged in a layered manner observed in the cross-sectional TEM image corresponds to a plane parallel to the a-b plane of the crystal.

Note that the crystal part is formed concurrently with deposition of the CAAC-OS film or is formed through crystallization treatment such as heat treatment. As described above, the c-axis of the crystal is aligned in a direction parallel to a normal vector of a formation surface or a normal vector of a top surface. Thus, for example, in the case where a shape of the CAAC-OS film is changed by etching or the like, the c-axis might not be necessarily parallel to a normal vector of a formation surface or a normal vector of a top surface of the CAAC-OS film.

Further, the degree of crystallinity in the CAAC-OS film is not necessarily uniform. For example, in the case where crystal growth leading to the CAAC-OS film occurs from the vicinity of the top surface of the film, the degree of the crystallinity in the vicinity of the top surface is higher than that in the vicinity of the formation surface in some cases. Further, when an impurity is added to the CAAC-OS film, the crystallinity in a region to which the impurity is added is changed, and the degree of crystallinity in the CAAC-OS film varies depending on regions.

Note that when the CAAC-OS film with an InGaZnO₄ crystal is analyzed by an out-of-plane method, a peak of 2θ may also be observed at around 36°, in addition to the peak of 2θ at around 31°. The peak of 2θ at around 36° indicates that a crystal having no c-axis alignment is included in part of the CAAC-OS film. It is preferable that in the CAAC-OS film, a peak of 2θ appear at around 31° and a peak of 2θ do not appear at around 36°.

The CAAC-OS film is an oxide semiconductor film having low impurity concentration. The impurity is an element other than the main components of the oxide semiconductor film, such as hydrogen, carbon, silicon, or a transition metal element. In particular, an element that has higher bonding strength to oxygen than a metal element included in the oxide semiconductor film, such as silicon, disturbs the atomic arrangement of the oxide semiconductor film by depriving the oxide semiconductor film of oxygen and causes a reduction in crystallinity. Further, a heavy metal such as iron or nickel, argon, carbon dioxide, or the like has a large atomic radius (or molecular radius), and thus disturbs the atomic arrangement of the oxide semiconductor film and causes a reduction in crystallinity when it is contained in the oxide semiconductor film. Note that the impurity contained in the oxide semiconductor film might serve as a carrier trap or a carrier generation source.

Further, the CAAC-OS film is an oxide semiconductor film having a low density of defect states. For example, an oxygen vacancy in the oxide semiconductor film serves as a carrier trap or a carrier generation source in some cases when hydrogen is captured therein.

The state in which the impurity concentration is low and the density of defect states is low (the number of oxygen vacancies is small) is referred to as a highly purified intrinsic state or a substantially highly purified intrinsic state. A highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor film has few carrier generation sources, and thus can have a low carrier density. Thus, a transistor including the oxide semiconductor film rarely has negative threshold voltage (is rarely normally on). The highly purified intrinsic or substantially highly purified intrinsic oxide semiconductor film has few carrier traps. Accordingly, the transistor including the oxide semiconductor film has small variations in electrical characteristics and high reliability. Electric charge trapped by the carrier traps in the oxide semiconductor film takes a long time to be released, and might behave like fixed electric charge. Thus, the transistor that includes the oxide semiconductor film

having high impurity concentration and a high density of defect states has unstable electrical characteristics in some cases.

With the use of the CAAC-OS film in a transistor, variations in the electrical characteristics of the transistor due to irradiation with visible light or ultraviolet light are small.

Next, a microcrystalline oxide semiconductor film is described.

In an image obtained with a TEM, crystal parts cannot be found clearly in the microcrystalline oxide semiconductor film in some cases. In most cases, the size of a crystal part in the microcrystalline oxide semiconductor film is greater than or equal to 1 nm and less than or equal to 100 nm, or greater than or equal to 1 nm and less than or equal to 10 nm. An oxide semiconductor film including nanocrystal (nc), which is a microcrystal with a size greater than or equal to 1 nm and less than or equal to 10 nm, or a size greater than or equal to 1 nm and less than or equal to 3 nm, is specifically referred to as a nanocrystalline oxide semiconductor (nc-OS) film. In an image of the nc-OS film obtained with a TEM, for example, a crystal grain cannot be observed clearly in some cases.

In the nc-OS film, a microscopic region (e.g., a region with a size greater than or equal to 1 nm and less than or equal to 10 nm, in particular, a region with a size greater than or equal to 1 nm and less than or equal to 3 nm) has a periodic atomic order. Further, there is no regularity of crystal orientation between different crystal parts in the nc-OS film; thus, the orientation of the whole film is not observed. Accordingly, in some cases, the nc-OS film cannot be distinguished from an amorphous oxide semiconductor film depending on an analysis method. For example, when the nc-OS film is subjected to structural analysis by an out-of-plane method with an XRD apparatus using an X-ray having a diameter larger than that of a crystal part, a peak which shows a crystal plane does not appear. Further, a halo pattern is observed in an electron diffraction pattern (also referred to as a selected-area electron diffraction pattern) of the nc-OS film obtained by using an electron beam having a probe diameter (e.g., larger than or equal to 50 nm) larger than the diameter of a crystal part. Meanwhile, spots are observed in a nanobeam electron diffraction pattern of the nc-OS film obtained by using an electron beam having a probe diameter (e.g., larger than or equal to 1 nm and smaller than or equal to 30 nm) close to, or smaller than or equal to the diameter of a crystal part. In some cases, in a nanobeam electron diffraction pattern of the nc-OS film, regions with high luminance in a circular (ring) pattern are observed. Further, in a nanobeam electron diffraction pattern of the nc-OS film, a plurality of spots are shown in a ring-like region in some cases.

Since an nc-OS film is an oxide semiconductor film having more regularity than an amorphous oxide semiconductor film, the nc-OS film has a lower density of defect states than the amorphous oxide semiconductor film. However, there is no regularity of crystal orientation between different crystal parts in the nc-OS film; hence, the nc-OS film has a higher density of defect states than a CAAC-OS film.

Note that an oxide semiconductor film may be a stacked film including two or more films of an amorphous oxide semiconductor film, a microcrystalline oxide semiconductor film, and a CAAC-OS film, for example.

A CAAC-OS film can be deposited by a sputtering method with a polycrystalline oxide semiconductor sputtering target, for example. When ions collide with the sputter-

ing target, a crystal region included in the sputtering target may be separated from the target along the a-b plane; in other words, a sputtered particle having a plane parallel to the a-b plane (a flat-plate-like sputtered particle or a pellet-like sputtered particle) might flake off from the target. In this case, the flat-plate-like sputtered particle or the pellet-like sputtered particle is electrically charged and thus reaches a substrate while maintaining its crystal state without being aggregated in plasma, whereby a CAAC-OS film can be formed.

In the case where the second oxide semiconductor layer **132** is formed using an In-M-Zn oxide (M is Ga, Y, Zr, La, Ce, or Nd) and a sputtering target whose atomic ratio of In to M and Zn is $a_1:b_1:c_1$ is used for forming the second oxide semiconductor layer **132**, a_1/b_1 is preferably greater than or equal to $1/3$ and less than or equal to 6, further preferably greater than or equal to 1 and less than or equal to 6, and c_1/b_1 is preferably greater than or equal to $1/3$ and less than or equal to 6, further preferably greater than or equal to 1 and less than or equal to 6. Note that when c_1/b_1 is greater than or equal to 1 and less than or equal to 6, a CAAC-OS film is easily formed as the second oxide semiconductor layer **132**. Typical examples of the atomic ratio of In to M and Zn of the target are 1:1:1, 3:1:2, and 5:5:6.

In the case where the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133** are each formed using an In-M-Zn oxide (M is Ga, Y, Zr, La, Ce, or Nd) and a sputtering target whose atomic ratio of In to M and Zn is $a_2:b_2:c_2$ is used for forming the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133**, a_2/b_2 is preferably less than a_1/b_1 , and c_2/b_2 is preferably greater than or equal to $1/3$ and less than or equal to 6, further preferably greater than or equal to 1 and less than or equal to 6. Note that when c_2/b_2 is greater than or equal to 1 and less than or equal to 6, CAAC-OS films are easily formed as the first oxide semiconductor layer **131** and the third oxide semiconductor layer **133**. Typical examples of the atomic ratio of In to M and Zn of the target are 1:3:2, 1:3:3, 1:3:4, and 1:3:6.

First heat treatment may be performed after the second oxide semiconductor layer **132** is formed. The first heat treatment may be performed at a temperature higher than or equal to 250° C. and lower than or equal to 650° C., preferably higher than or equal to 300° C. and lower than or equal to 500° C., in an inert gas atmosphere, in an atmosphere containing an oxidizing gas at 10 ppm or more, or under reduced pressure. Alternatively, the first heat treatment may be performed in such a manner that heat treatment is performed in an inert gas atmosphere, and then another heat treatment is performed in an atmosphere containing an oxidizing gas at 10 ppm or more in order to compensate desorbed oxygen. By the first heat treatment, the crystallinity of the second oxide semiconductor layer **132** can be improved, and in addition, impurities such as hydrogen and water can be removed from the base insulating film **120** and the first oxide semiconductor layer **131**. Note that the first heat treatment may be performed before etching for formation of the second oxide semiconductor layer **132**.

Next, a first conductive film to be the source electrode layer **140** and the drain electrode layer **150** is formed over the first oxide semiconductor layer **131** and the second oxide semiconductor layer **132**. For the first conductive film, Al, Cr, Cu, Ta, Ti, Mo, W, or an alloy material containing any of these as its main component can be used. For example, a 100-nm-thick titanium film is formed by a sputtering method or the like. Alternatively, a tungsten film may be formed by a CVD method.

25

Then, the first conductive film is etched so as to be divided over the second oxide semiconductor layer **132** to form the source electrode layer **140** and the drain electrode layer **150** (see FIG. 7C). At this time, the first conductive film may be over-etched, so that the second oxide semiconductor layer **132** is partly etched.

Subsequently, a third oxide semiconductor film **333** to be the third oxide semiconductor layer **133** is formed over the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, the source electrode layer **140**, and the drain electrode layer **150**. In the third oxide semiconductor film **333**, a microcrystalline layer is formed in the vicinity of the interface with the second oxide semiconductor layer **132**, and a crystalline layer in which c-axes are aligned is formed over the microcrystalline layer.

Note that second heat treatment may be performed after the third oxide semiconductor film **333** is formed. The second heat treatment can be performed under the conditions similar to those of the first heat treatment. The second heat treatment can remove impurities such as hydrogen and water from the third oxide semiconductor film **333**, the first oxide semiconductor layer **131**, and the second oxide semiconductor layer **132**.

Next, an insulating film **360** to be the gate insulating film **160** is formed over the third oxide semiconductor film **333**. The insulating film **360** can be formed using aluminum oxide, magnesium oxide, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, gallium oxide, germanium oxide, yttrium oxide, zirconium oxide, lanthanum oxide, neodymium oxide, hafnium oxide, tantalum oxide, or the like. The insulating film **360** may be a stack including any of the above materials. The insulating film **360** can be formed by a sputtering method, a CVD method, an MBE method, an ALD method, a PLD method, or the like.

Then, a second conductive film **370** to be the gate electrode layer **170** is formed over the insulating film **360** (see FIG. 8A). For the second conductive film **370**, Al, Ti, Cr, Co, Ni, Cu, Y, Zr, Mo, Ru, Ag, Ta, W, or an alloy material containing any of these as its main component can be used. The second conductive film **370** can be formed by a sputtering method, a CVD method, or the like. A stack including a conductive film containing any of the above materials and a conductive film containing nitrogen, or a conductive film containing nitrogen may be used for the second conductive film **370**.

After that, the second conductive film **370** is selectively etched using a resist mask to form the gate electrode layer **170**.

Then, the insulating film **360** is selectively etched using the resist mask or the gate electrode layer **170** as a mask to form the gate insulating film **160**.

Subsequently, the third oxide semiconductor film **333** is etched using the resist mask or the gate electrode layer **170** as a mask to form the third oxide semiconductor layer **133** (see FIG. 8B).

The second conductive film **370**, the insulating film **360**, and the third oxide semiconductor film **333** may be etched individually or successively. Note that either dry etching or wet etching may be used as the etching method, and an appropriate etching method may be selected individually.

Next, the insulating layer **180** and the insulating layer **185** are formed over the source electrode layer **140**, the drain electrode layer **150**, and the gate electrode layer **170** (see FIG. 8C). The insulating layer **180** and the insulating layer **185** can be formed using a material and a method which are

26

similar to those of the base insulating film **120**. Note that it is particularly preferable to use aluminum oxide for the insulating layer **180**.

Oxygen may be added to the insulating layer **180** by an ion implantation method, an ion doping method, a plasma immersion ion implantation method, or the like. Adding oxygen enables the insulating layer **180** to supply oxygen much easily to the oxide semiconductor layer **130**.

Next, third heat treatment may be performed. The third heat treatment can be performed under conditions similar to those of the first heat treatment. By the third heat treatment, excess oxygen is easily released from the base insulating film **120**, the gate insulating film **160**, and the insulating layer **180**, so that oxygen vacancies in the oxide semiconductor layer **130** can be reduced.

Through the above process, the transistor **100** illustrated in FIGS. 1A to 1C can be fabricated.

This embodiment can be combined as appropriate with any of the other embodiments and an example in this specification.

Embodiment 3

In this embodiment, an example of a semiconductor device (storage device) which includes the transistor of one embodiment of the present invention, which can retain stored data even when not powered, and which has an unlimited number of write cycles is described with reference to drawings.

FIG. 9A is a cross-sectional view of the semiconductor device, and FIG. 9B is a circuit diagram of the semiconductor device.

The semiconductor device illustrated in FIGS. 9A and 9B includes a transistor **3200** including a first semiconductor material in a lower portion, and a transistor **3300** including a second semiconductor material and a capacitor **3400** in an upper portion. Note that the transistor **100** described in Embodiment 1 can be used as the transistor **3300**.

One electrode of the capacitor **3400** is formed using the same material as a source electrode layer or a drain electrode layer of the transistor **3300**, the other electrode of the capacitor **3400** is formed using the same material as a gate electrode layer of the transistor **3300**, and a dielectric of the capacitor **3400** is formed using the same material as the gate insulating film **160** and the third oxide semiconductor layer **133** of the transistor **3300**; thus, the capacitor **3400** can be formed at the same time as the transistor **3300**.

Here, the first semiconductor material and the second semiconductor material preferably have different energy gaps. For example, the first semiconductor material may be a semiconductor material (such as silicon) other than an oxide semiconductor, and the second semiconductor material may be the oxide semiconductor described in Embodiment 1. A transistor including a material other than an oxide semiconductor can operate at high speed easily. In contrast, a transistor including an oxide semiconductor enables charge to be retained for a long time owing to its electrical characteristics, that is, the low off-state current.

Although both of the above transistors are n-channel transistors in the following description, it is needless to say that p-channel transistors can be used. The specific structure of the semiconductor device, such as a material used for the semiconductor device and the structure of the semiconductor device, needs not to be limited to that described here except for the use of the transistor described in Embodiment 1, which is formed using an oxide semiconductor, for retaining data.

The transistor 3200 in FIG. 9A includes a channel formation region provided in a substrate 3000 containing a semiconductor material (such as crystalline silicon), impurity regions provided such that the channel formation region is provided therebetween, intermetallic compound regions in contact with the impurity regions, a gate insulating film provided over the channel formation region, and a gate electrode layer provided over the gate insulating film. Note that a transistor whose source electrode layer and drain electrode layer are not illustrated in a drawing may also be referred to as a transistor for the sake of convenience. Further, in such a case, in description of a connection of a transistor, a source region and a source electrode layer may be collectively referred to as a source electrode layer, and a drain region and a drain electrode layer may be collectively referred to as a drain electrode layer. That is, in this specification, the term "source electrode layer" might include a source region.

An element isolation insulating layer 3100 is formed on the substrate 3000 so as to surround the transistor 3200, and an insulating layer 3150 is formed so as to cover the transistor 3200. Note that the element isolation insulating layer 3100 can be formed by an element isolation technique such as local oxidation of silicon (LOCOS) or shallow trench isolation (STI).

In the case where the transistor 3200 is formed using a crystalline silicon substrate, for example, the transistor 3200 can operate at high speed. Thus, when the transistor is used as a reading transistor, data can be read at high speed.

The transistor 3300 is provided over the insulating layer 3150, and the wiring electrically connected to the source electrode layer or the drain electrode layer of the transistor 3300 serves as the one electrode of the capacitor 3400. Further, the wiring is electrically connected to the gate electrode layer of the transistor 3200.

The transistor 3300 in FIG. 9A is a top-gate transistor in which a channel is formed in an oxide semiconductor layer. Since the off-state current of the transistor 3300 is low, stored data can be retained for a long period owing to such a transistor. In other words, refresh operation becomes unnecessary or the frequency of the refresh operation in a semiconductor storage device can be extremely low, which leads to a sufficient reduction in power consumption.

Further, an electrode 3250 is provided so as to overlap with the transistor 3300 with the insulating layer 3150 provided therebetween. By supplying an appropriate potential to the electrode 3250 and using the electrode 3250 as a second gate electrode, the threshold voltage of the transistor 3300 can be controlled. In addition, long-term reliability of the transistor 3300 can be improved. When the electrode operates with the same potential as that of the gate electrode of the transistor 3300, on-state current can be increased. Note that the electrode 3250 is not necessarily provided.

The transistor 3300 and the capacitor 3400 can be formed over the substrate over which the transistor 3200 is formed as illustrated in FIG. 9A, which enables the degree of the integration of the semiconductor device to be increased.

An example of a circuit configuration of the semiconductor device in FIG. 9A is illustrated in FIG. 9B.

In FIG. 9B, a first wiring 3001 is electrically connected to a source electrode layer of the transistor 3200. A second wiring 3002 is electrically connected to a drain electrode layer of the transistor 3200. A third wiring 3003 is electrically connected to one of the source electrode layer and the drain electrode layer of the transistor 3300. A fourth wiring 3004 is electrically connected to the gate electrode layer of the transistor 3300. The gate electrode layer of the transistor

3200 and the other of the source electrode layer and the drain electrode layer of the transistor 3300 are electrically connected to the one electrode of the capacitor 3400. A fifth wiring 3005 is electrically connected to the other electrode of the capacitor 3400. Note that a component corresponding to the electrode 3250 is not illustrated.

The semiconductor device in FIG. 9B utilizes a feature that the potential of the gate electrode layer of the transistor 3200 can be retained, and thus enables writing, retaining, and reading of data as follows.

Writing and retaining of data are described. First, the potential of the fourth wiring 3004 is set to a potential at which the transistor 3300 is turned on, so that the transistor 3300 is turned on. Accordingly, the potential of the third wiring 3003 is supplied to the gate electrode layer of the transistor 3200 and the capacitor 3400. That is, a predetermined charge is supplied to the gate electrode layer of the transistor 3200 (writing). Here, one of two kinds of charges providing different potential levels (hereinafter referred to as a low-level charge and a high-level charge) is supplied. After that, the potential of the fourth wiring 3004 is set to a potential at which the transistor 3300 is turned off, so that the transistor 3300 is turned off. Thus, the charge supplied to the gate electrode layer of the transistor 3200 is retained (retaining).

Since the off-state current of the transistor 3300 is extremely low, the charge of the gate electrode layer of the transistor 3200 is retained for a long time.

Next, reading of data is described. An appropriate potential (a reading potential) is supplied to the fifth wiring 3005 while a predetermined potential (a constant potential) is supplied to the first wiring 3001, whereby the potential of the second wiring 3002 varies depending on the amount of charge retained in the gate electrode layer of the transistor 3200. This is because in general, in the case of using an n-channel transistor as the transistor 3200, an apparent threshold voltage V_{th_H} at the time when the high-level charge is given to the gate electrode layer of the transistor 3200 is lower than an apparent threshold voltage V_{th_L} at the time when the low-level charge is given to the gate electrode layer of the transistor 3200. Here, an apparent threshold voltage refers to the potential of the fifth wiring 3005 which is needed to turn on the transistor 3200. Thus, the potential of the fifth wiring 3005 is set to a potential V_0 which is between V_{th_H} and V_{th_L} , whereby charge supplied to the gate electrode layer of the transistor 3200 can be determined. For example, in the case where the high-level charge is supplied in writing and the potential of the fifth wiring 3005 is $V_0(>V_{th_H})$, the transistor 3200 is turned on. In the case where the low-level charge is supplied in writing, even when the potential of the fifth wiring 3005 is $V_0(<V_{th_L})$, the transistor 3200 remains off. Thus, the data retained in the gate electrode layer can be read by determining the potential of the second wiring 3002.

Note that in the case where memory cells are arrayed, it is necessary that only data of a desired memory cell be able to be read. The fifth wiring 3005 in the case where data is not read may be supplied with a potential at which the transistor 3200 is turned off regardless of the state of the gate electrode layer, that is, a potential lower than V_{th_H} . Alternatively, the fifth wiring 3005 may be supplied with a potential at which the transistor 3200 is turned on regardless of the state of the gate electrode layer, that is, a potential higher than V_{th_L} .

When including a transistor having a channel formation region formed using an oxide semiconductor and having an extremely low off-state current, the semiconductor device described in this embodiment can retain stored data for an

extremely long time. In other words, refresh operation becomes unnecessary or the frequency of the refresh operation can be extremely low, which leads to a sufficient reduction in power consumption. Moreover, stored data can be retained for a long time even when power is not supplied (note that a potential is preferably fixed).

Further, in the semiconductor device described in this embodiment, high voltage is not needed for writing data and there is no problem of deterioration of elements. Unlike in a conventional nonvolatile memory, for example, it is not necessary to inject and extract electrons into and from a floating gate; thus, a problem such as deterioration of a gate insulating film is unlikely to be caused. That is, the semiconductor device of the disclosed invention does not have a limit on the number of times data can be rewritten, which is a problem of a conventional nonvolatile memory, and the reliability thereof is drastically improved. Furthermore, data is written depending on the state of the transistor (on or off), whereby high-speed operation can be easily achieved.

As described above, a miniaturized and highly integrated semiconductor device having high electrical characteristics can be provided.

This embodiment can be combined as appropriate with any of the other embodiments and an example in this specification.

Embodiment 4

In this embodiment, a semiconductor device including the transistor of one embodiment of the present invention, which can retain stored data even when not powered, which does not have a limit on the number of write cycles, and which has a structure different from that described in Embodiment 3, is described.

FIG. 10 illustrates an example of a circuit configuration of the semiconductor device. In the semiconductor device, a first wiring 4500 is electrically connected to a source electrode layer of a transistor 4300, a second wiring 4600 is electrically connected to a gate electrode layer of the transistor 4300, and a drain electrode layer of the transistor 4300 is electrically connected to a first terminal of a capacitor 4400. Note that the transistor 100 described in Embodiment 1 can be used as the transistor 4300 included in the semiconductor device. The first wiring 4500 can serve as a bit line and the second wiring 4600 can serve as a word line.

The semiconductor device (a memory cell 4250) can have a connection mode similar to that of the transistor 3300 and the capacitor 3400 illustrated in FIGS. 9A and 9B. Thus, the capacitor 4400 can be formed in the same process and at the same time as the transistor 4300 in a manner similar to that of the capacitor 3400 described in Embodiment 3.

Next, writing and retaining of data in the semiconductor device (the memory cell 4250) illustrated in FIG. 10 are described.

First, a potential at which the transistor 4300 is turned on is supplied to the second wiring 4600, so that the transistor 4300 is turned on. Accordingly, the potential of the first wiring 4500 is supplied to the first terminal of the capacitor 4400 (writing). After that, the potential of the second wiring 4600 is set to a potential at which the transistor 4300 is turned off, so that the transistor 4300 is turned off. Thus, the potential of the first terminal of the capacitor 4400 is retained (retaining).

The transistor 4300 including an oxide semiconductor has an extremely low off-state current. For that reason, the potential of the first terminal of the capacitor 4400 (or a

charge accumulated in the capacitor 4400) can be retained for an extremely long time by turning off the transistor 4300.

Next, reading of data is described. When the transistor 4300 is turned on, the first wiring 4500 which is in a floating state and the capacitor 4400 are electrically connected to each other, and the charge is redistributed between the first wiring 4500 and the capacitor 4400. As a result, the potential of the first wiring 4500 is changed. The amount of change in potential of the first wiring 4500 varies depending on the potential of the first terminal of the capacitor 4400 (or the charge accumulated in the capacitor 4400).

For example, the potential of the first wiring 4500 after the charge redistribution is $(C_B \times V_{B0} + C \times V)/(C_B + C)$, where V is the potential of the first terminal of the capacitor 4400, C is the capacitance of the capacitor 4400, C_B is the capacitance component of the first wiring 4500, and V_{B0} is the potential of the first wiring 4500 before the charge redistribution. Thus, it can be found that, assuming that the memory cell 4250 is in either of two states in which the potential of the first terminal of the capacitor 4400 is V_1 and V_0 ($V_1 > V_0$), the potential of the first wiring 4500 in the case of retaining the potential V_1 ($= (C_B \times V_{B0} + C \times V_1)/(C_B + C)$) is higher than the potential of the first wiring 4500 in the case of retaining the potential V_0 ($= (C_B \times V_{B0} + C \times V_0)/(C_B + C)$).

Then, by comparing the potential of the first wiring 4500 with a predetermined potential, data can be read.

As described above, the semiconductor device (the memory cell 4250) illustrated in FIG. 10 can retain charge that is accumulated in the capacitor 4400 for a long time because the off-state current of the transistor 4300 is extremely low. In other words, refresh operation becomes unnecessary or the frequency of the refresh operation can be extremely low, which leads to a sufficient reduction in power consumption. Moreover, stored data can be retained for a long time even when power is not supplied.

A substrate over which a driver circuit for the memory cell 4250 is formed and the memory cell 4250 illustrated in FIG. 10 are preferably stacked. When the memory cell 4250 and the driver circuit are stacked, the size of the semiconductor device can be reduced. Note that there is no limitation on the numbers of the memory cells 4250 and the driver circuits which are stacked.

It is preferable that a semiconductor material of a transistor included in the driver circuit be different from that of the transistor 4300. For example, silicon, germanium, silicon germanium, silicon carbide, or gallium arsenide can be used, and a single crystal semiconductor is preferably used. A transistor formed using such a semiconductor material can operate at higher speed than a transistor formed using an oxide semiconductor and is suitable for the driver circuit for the memory cell 4250.

As described above, a miniaturized and highly integrated semiconductor device having high electrical characteristics can be provided.

This embodiment can be combined as appropriate with any of the other embodiments and an example in this specification.

Embodiment 5

In this embodiment, an example of a circuit including the transistor of one embodiment of the present invention will be described with reference to the drawings.

FIG. 11A is a circuit diagram of a semiconductor device and FIGS. 11C and 11D are each a cross-sectional view of a semiconductor device. FIGS. 11C and 11D each illustrate a cross-sectional view of a transistor 2100 in a channel

31

length direction on the left and a cross-sectional view of the transistor **2100** in a channel width direction on the right. In the circuit diagram, "OS" is written beside a transistor in order to clearly demonstrate that the transistor includes an oxide semiconductor.

The semiconductor devices illustrated in FIGS. **11C** and **11D** each include a transistor **2200** containing a first semiconductor material in a lower portion and the transistor **2100** containing a second semiconductor material in an upper portion. Here, an example is described in which the transistor **100** described in Embodiment 1 as an example is used as the transistor **2100** containing the second semiconductor material.

Here, the first semiconductor material and the second semiconductor material preferably have different energy gaps. For example, the first semiconductor material may be a semiconductor material (e.g., silicon, germanium, silicon germanium, silicon carbide, or gallium arsenic) other than an oxide semiconductor, and the second semiconductor material may be the oxide semiconductor described in Embodiment 1. A transistor including single crystal silicon or the like as a material other than an oxide semiconductor can operate at high speed easily. In contrast, a transistor including an oxide semiconductor has the low off-state current.

Although the transistor **2200** is a p-channel transistor here, it is needless to say that an n-channel transistor can be used to form a circuit having a different configuration. The specific structure of the semiconductor device, such as a material used for the semiconductor device and the structure of the semiconductor device, needs not to be limited to that described here except for the use of the transistor described in Embodiment 1, which is formed using an oxide semiconductor.

FIGS. **11A**, **11C**, and **11D** each illustrate a configuration example of what is called a CMOS circuit, in which a p-channel transistor and an n-channel transistor are connected in series and gates of the transistors are connected.

The circuit can operate at high speed because the transistor of one embodiment of the present invention including an oxide semiconductor has high on-state current.

FIG. **11C** illustrates a configuration in which the transistor **2100** is provided over the transistor **2200** with an insulating layer **2201** provided therebetween. Further, a plurality of wirings **2202** are provided between the transistor **2200** and the transistor **2100**. Furthermore, wirings and electrodes provided in the upper portion and the lower portion are electrically connected to each other through a plurality of plugs **2203** embedded in insulating layers. Note that an insulating layer **2204** covering the transistor **2100**, a wiring **2205** over the insulating layer **2204**, and a wiring **2206** formed by processing a conductive film that is also used for a pair of electrodes of the transistor are provided.

When two transistors are stacked as described above, the area occupied by the circuit can be reduced and a plurality of circuits can be arranged with higher density.

In FIG. **11C**, one of a source and a drain of the transistor **2100** is electrically connected to one of a source and a drain of the transistor **2200** through the wirings **2202** and the plugs **2203**. Further, the gate of the transistor **2100** is electrically connected to the gate of the transistor **2200** through the wiring **2205**, the wiring **2206**, the plugs **2203**, the wiring **2202**, and the like.

In the configuration illustrated in FIG. **11D**, an opening portion in which the plug **2203** is embedded is provided in a gate insulating film of the transistor **2100**, and the gate of the transistor **2100** is in contact with the plug **2203** through

32

the opening portion. Such a configuration makes it possible to achieve the integration of the circuit easily and to reduce the lengths and the number of wirings and plugs used to be smaller than those in the configuration illustrated in FIG. **11C**; thus, the circuit can operate at higher speed.

Note that when a connection between the electrodes of the transistor **2100** and the transistor **2200** is changed from that in the configuration illustrated in FIG. **11C** or FIG. **11D**, a variety of circuits can be formed. For example, a circuit having a configuration in which a source and a drain of a transistor are connected to those of another transistor as illustrated in FIG. **11B** can operate as what is called an analog switch.

This embodiment can be combined as appropriate with any of the other embodiments and an example in this specification.

Embodiment 6

In this embodiment, a semiconductor device which includes the transistor of one embodiment of the present invention and has an image sensor function for reading data of an object will be described.

FIG. **12** illustrates an example of an equivalent circuit of a semiconductor device having an image sensor function.

In a photodiode **610**, one electrode is electrically connected to a photodiode reset signal line **661**, and the other electrode is electrically connected to a gate of a transistor **640**. One of a source and a drain of the transistor **640** is electrically connected to a photosensor reference signal line **672**, and the other of the source and the drain thereof is electrically connected to one of a source and a drain of a transistor **650**. A gate of the transistor **650** is electrically connected to a gate signal line **662**, and the other of the source and the drain thereof is electrically connected to a photosensor output signal line **671**.

As the photodiode **610**, for example, a pin photodiode in which a semiconductor layer having p-type conductivity, a high-resistance semiconductor layer (semiconductor layer having i-type conductivity), and a semiconductor layer having n-type conductivity are stacked can be used.

With detection of light that enters the photodiode **610**, data of an object can be read. Note that a light source such as a backlight can be used at the time of reading data of an object.

Note that as each of the transistor **640** and the transistor **650**, the transistor **100** described in Embodiment 1 in which a channel is formed in an oxide semiconductor can be used. In FIG. **12**, "OS" is written beside each of the transistor **640** and the transistor **650** so that it can be clearly identified that the transistors include an oxide semiconductor. The transistor **640** and the transistor **650** are electrically stable transistors that have high on-state current and less change in electrical characteristics. With the transistor, the semiconductor device having an image sensor function, which is illustrated in FIG. **12**, can be highly reliable.

This embodiment can be combined as appropriate with any of the other embodiments and an example in this specification.

Embodiment 7

The transistor described in Embodiments 1 and 2 can be used in a semiconductor device such as a display device, a storage device, a CPU, a digital signal processor (DSP), an LSI such as a custom LSI or a programmable logic device (PLD), a radio frequency identification (RF-ID), an inverter,

or an image sensor. In this embodiment, electronic devices each including the semiconductor device will be described.

Examples of the electronic devices having the semiconductor devices include display devices of televisions, monitors, and the like, lighting devices, personal computers, word processors, image reproduction devices, portable audio players, radios, tape recorders, stereos, phones, cordless phones, mobile phones, car phones, transceivers, wireless devices, game machines, calculators, portable information terminals, electronic notebooks, e-book readers, electronic translators, audio input devices, video cameras, digital still cameras, electric shavers, IC chips, high-frequency heating appliances such as microwave ovens, electric rice cookers, electric washing machines, electric vacuum cleaners, air-conditioning systems such as air conditioners, dishwashers, dish dryers, clothes dryers, futon dryers, electric refrigerators, electric freezers, electric refrigerator-freezers, freezers for preserving DNA, radiation counters, and medical equipment such as dialyzers and X-ray diagnostic equipment. In addition, the examples of the electronic devices include alarm devices such as smoke detectors, heat detectors, gas alarm devices, and security alarm devices. Further, the examples of the electronic devices also include industrial equipment such as guide lights, traffic lights, belt conveyors, elevators, escalators, industrial robots, and power storage systems. In addition, moving objects and the like driven by fuel engines and electric motors using power from non-aqueous secondary batteries are also included in the category of electronic devices. Examples of the moving objects include electric vehicles (EV), hybrid electric vehicles (HEV) which include both an internal-combustion engine and a motor, plug-in hybrid electric vehicles (PHEV), tracked vehicles in which caterpillar tracks are substituted for wheels of these vehicles, motorized bicycles including motor-assisted bicycles, motorcycles, electric wheelchairs, golf carts, boats or ships, submarines, helicopters, aircrafts, rockets, artificial satellites, space probes, planetary probes, and spacecrafts. Some specific examples of these electronic devices are illustrated in FIGS. 13A to 13C.

In a television set **8000** illustrated in FIG. 13A, a display portion **8002** is incorporated in a housing **8001**. The display portion **8002** can display an image and a speaker portion **8003** can output sound. A storage device including the transistor of one embodiment of the present invention can be used for a driver circuit for operating the display portion **8002**.

The television set **8000** may also include a CPU **8004** for performing information communication or a memory. For the CPU **8004** and the memory, a CPU or a storage device including the transistor of one embodiment of the present invention can be used.

An alarm device **8100** illustrated in FIG. 13A is a residential fire alarm, which is an example of an electronic device including a sensor portion **8102** for smoke or heat and a microcomputer **8101**. Note that the microcomputer **8101** includes a storage device or a CPU including the transistor of one embodiment of the present invention.

An air conditioner which includes an indoor unit **8200** and an outdoor unit **8204** illustrated in FIG. 13A is an example of an electronic device including the transistor, the storage device, the CPU, or the like described in any of the above embodiments. Specifically, the indoor unit **8200** includes a housing **8201**, an air outlet **8202**, a CPU **8203**, and the like. Although the CPU **8203** is provided in the indoor unit **8200** in FIG. 13A, the CPU **8203** may be provided in the outdoor unit **8204**. Alternatively, the CPU **8203** may be provided in both the indoor unit **8200** and the outdoor unit **8204**. By

using any of the transistors of one embodiment of the present invention for the CPU in the air conditioner, a reduction in power consumption of the air conditioner can be achieved.

An electric refrigerator-freezer **8300** illustrated in FIG. 13A is an example of an electronic device including the transistor, the storage device, the CPU, or the like described in any of the above embodiments. Specifically, the electric refrigerator-freezer **8300** includes a housing **8301**, a door for a refrigerator **8302**, a door for a freezer **8303**, a CPU **8304**, and the like. In FIG. 13A, the CPU **8304** is provided in the housing **8301**. When the transistor of one embodiment of the present invention is used for the CPU **8304** of the electric refrigerator-freezer **8300**, a reduction in power consumption of the electric refrigerator-freezer **8300** can be achieved.

FIGS. 13B and 13C illustrate an example of an electric vehicle which is an example of an electronic device. An electric vehicle **9700** is equipped with a secondary battery **9701**. The output of the electric power of the secondary battery **9701** is adjusted by a circuit **9702** and the electric power is supplied to a driving device **9703**. The circuit **9702** is controlled by a processing unit **9704** including a ROM, a RAM, a CPU, or the like which is not illustrated. When the transistor of one embodiment of the present invention is used for the CPU in the electric vehicle **9700**, a reduction in power consumption of the electric vehicle **9700** can be achieved.

The driving device **9703** includes a DC motor or an AC motor either alone or in combination with an internal-combustion engine. The processing unit **9704** outputs a control signal to the circuit **9702** on the basis of input data such as data of operation (e.g., acceleration, deceleration, or stop) by a driver or data during driving (e.g., data on an upgrade or a downgrade, or data on a load on a driving wheel) of the electric vehicle **9700**. The circuit **9702** adjusts the electric energy supplied from the secondary battery **9701** in accordance with the control signal of the processing unit **9704** to control the output of the driving device **9703**. In the case where the AC motor is mounted, although not illustrated, an inverter which converts a direct current into an alternate current is also incorporated.

This embodiment can be combined as appropriate with any of the other embodiments and an example in this specification.

EXAMPLE

In this example, observation results of the stack including oxide semiconductor layers described in Embodiment 1 will be described in detail.

FIG. 14 is a cross-sectional view illustrating a structure of a sample used in this example. The sample includes a base insulating film **420** over a substrate **410**, a stack including a first oxide semiconductor layer **431** and a second oxide semiconductor layer **432** over the base insulating film, and a third oxide semiconductor layer **433** formed over the stack. Note that the first oxide semiconductor layer **431**, the second oxide semiconductor layer **432**, and the third oxide semiconductor layer **433** correspond to the first oxide semiconductor layer **131**, the second oxide semiconductor layer **132**, and the third oxide semiconductor layer **133** described in Embodiment 1, respectively.

Here, a method for forming the sample illustrated in FIG. 14 is described.

First, a silicon wafer was used as the substrate **410**, and the silicon wafer was subjected to thermal oxidation to form a silicon oxide film serving as the base insulating film **420**.

Next, a first In—Ga—Zn oxide film whose atomic ratio of In to Ga and Zn is 1:3:4 and a second In—Ga—Zn oxide film whose atomic ratio of In to Ga and Zn is 1:1:1 were successively formed over the base insulating film 420 by a sputtering method. Note that the thickness of the first In—Ga—Zn oxide film and the thickness of the second In—Ga—Zn oxide film were 20 nm and 15 nm, respectively.

The first In—Ga—Zn oxide film was formed under the following conditions: an In—Ga—Zn oxide whose diameter is 8 inches and whose atomic ratio of In to Ga and Zn is 1:3:4 was used as a target, a sputtering gas containing argon and oxygen at a flow rate of 2:1 was used, the deposition pressure was 0.4 Pa, the electric power (DC) of 0.5 kW was supplied, the distance between the target and the substrate was 60 mm, and the substrate temperature was 200° C.

The second In—Ga—Zn oxide film was formed under the following conditions: an In—Ga—Zn oxide whose diameter is 8 inches and whose atomic ratio of In to Ga and Zn is 1:1:1 was used as a target, a sputtering gas containing argon and oxygen at a flow rate of 2:1 was used, the deposition pressure was 0.4 Pa, the electric power (DC) of 0.5 kW was supplied, the distance between the target and the substrate was 60 mm, and the substrate temperature was 300° C.

Then, the first In—Ga—Zn oxide film and the second In—Ga—Zn oxide film were subjected to heat treatment at 450° C. in a nitrogen atmosphere for one hour, and then subjected to heat treatment at 450° C. in an oxygen atmosphere for one hour.

After that, a 5-nm-thick tungsten film and a 20-nm-thick organic resin were formed over the second In—Ga—Zn oxide film, and a resist mask was formed by electron beam exposure.

Then, the organic resin and the tungsten film were selectively etched using the resist mask. As the etching, two steps of etching were performed using an inductively coupled plasma dry etching apparatus.

The first step of etching was performed under the following conditions: 100% carbon tetrafluoride was used as an etching gas, the pressure was 0.67 Pa, the electric power of 2000 W was supplied, the bias power was 50 W, the substrate temperature was -10° C., and the etching time was 12 seconds. The second step of etching was performed under the following conditions: an etching gas containing carbon tetrafluoride and oxygen at a flow rate of 3:2 was used, the pressure was 2.0 Pa, the electric power of 1000 W was supplied, the substrate bias power was 25 W, the substrate temperature was -10° C., and the etching time was 8 seconds.

Next, the first In—Ga—Zn oxide film and the second In—Ga—Zn oxide film were selectively etched using the organic resin and the tungsten film as a mask, so that a stack including the first oxide semiconductor layer 431 and the second oxide semiconductor layer 432 was formed. The etching was performed under the following conditions: an inductively coupled plasma dry etching apparatus was used, an etching gas containing methane and argon at a flow rate of 1:2 was used, the pressure was 1.0 Pa, the electric power of 600 W was supplied, the substrate bias power was 100 W, the substrate temperature was 70° C., and the etching time was 82 seconds.

After that, the organic resin and the tungsten film were etched under the following conditions: an inductively coupled plasma dry etching apparatus was used, an etching gas containing carbon tetrafluoride and oxygen at a flow rate of 3:2 was used, the pressure was 2.0 Pa, the electric power

of 1000 W was supplied, the substrate bias power was 25 W, the substrate temperature was -10° C., and the etching time was 6 seconds.

Then, the third oxide semiconductor layer 433 was formed to have a thickness of 10 nm over the stack including the first oxide semiconductor layer 431 and the second oxide semiconductor layer 432 by a sputtering method.

The third oxide semiconductor layer 433 was formed under the following conditions: an In—Ga—Zn oxide whose diameter is 8 inches and whose atomic ratio of In to Ga and Zn is 1:3:4 was used as a target, a sputtering gas containing argon and oxygen at a flow rate of 2:1 was used, the deposition pressure was 0.4 Pa, the electric power (DC) of 0.5 kW was supplied, the distance between the target and the substrate was 60 mm, and the substrate temperature was 200° C.

FIG. 15A is a cross-sectional TEM image of a region surrounded by a dotted line in FIG. 14. Although a crystal lattice is not observed in a region of several nanometers in the first oxide semiconductor layer 431 on the base insulating film 420 side, lattice fringes are observed in an upper portion of the region. Further, in the second oxide semiconductor layer 432, lattice fringes that are similar to those in the first oxide semiconductor layer 431 are observed. This means that the most part of the first oxide semiconductor layer 431 and the whole second oxide semiconductor layer 432 are formed of crystalline layers, and the directions of the lattice fringes demonstrate that the crystalline layers are each a CAAC-OS film in which c-axes are aligned in the direction perpendicular to its deposition surface.

In addition, although a crystal lattice is not observed in a region of several nanometers in the third oxide semiconductor layer 433 on the first oxide semiconductor layer 431 side or on the second oxide semiconductor layer 432 side, lattice fringes are observed in an upper portion of the region. This indicates that the third oxide semiconductor layer 433 includes a microcrystalline layer 433a and a crystalline layer 433b.

The lattice fringes in the crystalline layer 433b have different directions in a region over the second oxide semiconductor layer 432 and in a region that is formed to face a side surface of the first oxide semiconductor layer 431 or the second oxide semiconductor layer 432, which indicates that the crystalline layer 433b is a CAAC-OS film in which c-axes are aligned in the direction perpendicular to its deposition surface.

Further, as apparent from FIG. 15B, which is an enlarged view of a region surrounded by dotted lines in FIG. 15A, over a curved surface of an edge portion of the second oxide semiconductor layer 432, crystal fringes of the crystalline layer 433b in which c-axes are aligned in a direction perpendicular to the curved surface, with the microcrystalline layer 433a provided therebetween, are observed.

The above results of this example indicate that the stack including the oxide semiconductor layers, which is one embodiment of the present invention, can be formed.

This embodiment can be combined as appropriate with any of the other embodiments in this specification.

This application is based on Japanese Patent Application serial no. 2013-106337 filed with Japan Patent Office on May 20, 2013, the entire contents of which are hereby incorporated by reference.

What is claimed is:

1. A semiconductor device comprising:

- a first oxide semiconductor layer;
- a second oxide semiconductor layer over and in contact with the first oxide semiconductor layer;

37

a third oxide semiconductor layer over and in contact with the second oxide semiconductor layer;
 a gate insulating film over the third oxide semiconductor layer; and
 a gate electrode layer over the gate insulating film, 5
 wherein an indium content of the second oxide semiconductor layer is higher than an indium content of the first oxide semiconductor layer,
 wherein the indium content of the second oxide semiconductor layer is higher than an indium content of the 10
 third oxide semiconductor layer,
 wherein the third oxide semiconductor layer comprises a first region, a second region, and a third region,
 wherein the second region is positioned between the first 15
 region and the third region,
 wherein a top surface of the second region is lower than a top surface of the first region, and
 wherein a top surface of the second region is lower than a top surface of the third region. 20

2. The semiconductor device according to claim 1,
 wherein a conduction band minimum of the second oxide semiconductor layer is lower than a conduction band minimum of the first oxide semiconductor layer, and
 wherein the conduction band minimum of the second 25
 oxide semiconductor layer is lower than a conduction band minimum of the third oxide semiconductor layer.

3. A semiconductor device comprising:
 a first oxide semiconductor layer;
 a second oxide semiconductor layer over and in contact 30
 with the first oxide semiconductor layer;
 a third oxide semiconductor layer over and in contact with the second oxide semiconductor layer;
 a gate insulating film over the third oxide semiconductor 35
 layer; and
 a gate electrode layer over the gate insulating film,
 wherein an indium content of the second oxide semiconductor layer is higher than an indium content of the first oxide semiconductor layer,

38

wherein the indium content of the second oxide semiconductor layer is higher than an indium content of the third oxide semiconductor layer,
 wherein the third oxide semiconductor layer comprises a first layer and a second layer over the first layer, and wherein the first layer and the second layer have different crystallinities.

4. The semiconductor device according to claim 3,
 wherein a conduction band minimum of the second oxide semiconductor layer is lower than a conduction band minimum of the first oxide semiconductor layer, and wherein the conduction band minimum of the second oxide semiconductor layer is lower than a conduction band minimum of the third oxide semiconductor layer.

5. A semiconductor device comprising:
 a first oxide semiconductor layer;
 a second oxide semiconductor layer over and in contact with the first oxide semiconductor layer; and
 a third oxide semiconductor layer over and in contact with the second oxide semiconductor layer,
 wherein an indium content of the second oxide semiconductor layer is higher than an indium content of the first oxide semiconductor layer, and
 wherein the indium content of the second oxide semiconductor layer is higher than an indium content of the third oxide semiconductor layer.

6. The semiconductor device according to claim 5,
 wherein a gate insulating film is positioned over and in contact with the third region of the third oxide semiconductor layer, and
 wherein a gate electrode layer is positioned over the gate insulating film.

7. The semiconductor device according to claim 2,
 wherein a conduction band minimum of the second oxide semiconductor layer is lower than a conduction band minimum of the first oxide semiconductor layer, and wherein the conduction band minimum of the second oxide semiconductor layer is lower than a conduction band minimum of the third oxide semiconductor layer.

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